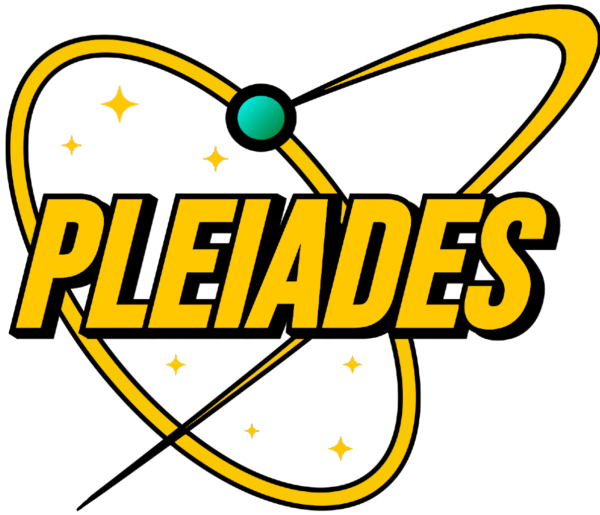




PLEIADES

Payload Launch & Experimental Instrumentation for Advanced Deployment & Exploration System

Critical Design Evaluation for PLEIADES with Payload Deployment

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February 23rd, 2025

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Mechanical and Aerospace Engineering & Electrical and Computer Engineering

The goal of the M5 - Critical Design Evaluation report is to comprehensively identify and expand upon potential failure modes and effects while demonstrating the team's progress in modeling and analysis. This report also outlines the development plan for the system and emphasizes the critical role of component prototyping and testing.

Failure Modes and Effects Analysis

Item / Function	Potential Failure Mode(s)	Potential Effect(s) of Failure	S e v	Potential Cause(s)/Mechanism(s) of Failure	P r o b	Current Design Controls	D e t	R P N	Recommended Action(s)
Propellant Grains	Failed/improper ignition, incomplete combustion	Rocket does not reach intended altitude	7	Poor mixing from manufacturer or Improper care of grains (i.e. proper storage, no moisture)	2	None	10	140	Focus on storing grains properly (according to manufacturer specifications)
Motor casing	Cracks in wall	Damage to rocket and surroundings	10	Improper material selection or excessive stress or manufacturing defect	1	Visual Inspection	2	20	Regularly check integrity and replace if needed
Motor casing	Bolt shear	Damage to rocket and surroundings	10	Improper material selection or excessive stress/fatigue or manufacturing defect or stripped threads	1	None	10	100	Run detailed simulations
Motor casing	Bolt tearout	Damage to rocket and surroundings	10	Improper material selection or excessive stress/fatigue or manufacturing defect	1	None	10	100	Run detailed simulations
Motor casing	Thermal deformation	Reduction in structural integrity, potential damage to rocket	9	Improper material selection or excessive stress or manufacturing defect	1	None	10	90	Ensure phenolic liner is intact
Nozzle Cap	Cracks in cap	Grains fall out or motor does not produce the intended thrust	3	Improper material selection or excessive stress or manufacturing defect	3	Visual Inspection	1	9	Be cautious while handling motor
Nozzle Cap	Cap falls off during storage	Grains fall out or motor does not produce the intended thrust	3	Improper fit or manufacturing defect	2	Visual Inspection	1	6	Ensure cap is properly attached
Aft Closure	Tearout	Decrease in thrust	7	Improper material selection or excessive stress/fatigue or manufacturing defect or stripped threads	1	None	10	70	Run detailed simulations
Aft Closure	Seal is not sufficient	Shifting of lining within casing	4	Improper seating	2	Visual Inspection	1	8	Ensure aft closure is properly attached
Aft Closure	Cracks/Damage	propellant/thrust leakage	4	Improper material selection or excessive stress or manufacturing defect	2	Visual Inspection	2	16	Check integrity and replace if needed
Nozzle	Broken nozzle	Loss of thrust and potential misfire	4	Improper material selection or excessive stress or manufacturing defect	2	Visual Inspection	1	8	Check integrity and replace if needed
Motor O-rings	O-ring snaps	Improper sealing (leading to leaks, higher temps, etc)	9	Environment/weather	3	Visual Inspection	4	108	Check integrity and replace if needed
Motor O-rings	Cracking	Improper sealing	9	Improper material selection or excessive stress or manufacturing defect or too old/stiff	1	Visual Inspection	2	18	Check integrity and replace if needed

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Aft Insulator	Damaged insulation	Damage to aft o-ring and other adjacent components	5	Manufacturing defect	1	None	10	50	None
Phenolic Liner	Holes in the lining	Leaks of propellant grain, decreasing thrust	9	Manufacturing defect	1	None	10	90	Proper storage to avoid damage
Forward Seal Disc	Damaged disk	Damage to the upper sections of the motor, possible combustion of the ejection charge	5	Manufacturing defect	1	None	10	50	None
Forward Seal Disc	Seal is not sufficient	Damage to the upper sections of the motor, possible combustion of the ejection charge	4	Improper seating	1	None	10	40	None
Aft Delay Spacer	Incorrect Size	Delay charge could move around, incorrect ignition	3	Assembly defect	1	Visual Inspection	1	3	If team changes delay, ensure spacer reflects the change
Delay insulator	Damaged insulation	Damage to delay element and other adjacent components, possible early ignition of the ejection charge	2	Manufacturing defect	1	None	10	20	None
Delay element	Failed/improper ignition	Parachute deploys at an incorrect time or not at all	10	Poor mixing from manufacturer or delay spacer is the incorrect size	2	None	10	200	None
Delay element	Element burns at an unexpected rate	early/late ignition of the ejection charge	7	Poor mixing from manufacturer or delay spacer is the incorrect size	2	None	10	140	None
Forward Closure	Tearout	Shifting of lining within casing	7	Improper material selection or excessive stress/fatigue or manufacturing defect or stripped threads	2	None	10	140	Run detailed simulations
Forward Closure	Seal is not sufficient	Shifting of lining within casing	4	Improper seating	2	Visual Inspection	1	8	Ensure forward closure is properly attached
Forward Closure	Cracks/Damage	delay element propellant leakage	4	Improper material selection or excessive stress or manufacturing defect	2	Visual Inspection	2	16	Check integrity and replace if needed
Forward Delay Spacer	Incorrect Size	Delay charge could move around, incorrect ignition	3	Assembly defect	2	Visual Inspection	1	6	If team changes delay, ensure spacer reflects the change
Ejection Charge	Failed/improper ignition, incomplete combustion	Parachute deploys at an incorrect time or not at all	10	Poor mixing from manufacturer or Improper spacing from fuse or Improper care of charge (i.e. proper storage, no moisture)	2	None	10	200	Focus on storing grains properly (according to manufacturer specifications)
Ejection Charge	Element burns at an unexpected rate	early/late ignition of the ejection charge	7	Manufacturing defect	1	None	10	70	None

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Ejection Charge	Too much charge	More force than expected can damage electronics & recovery system	4	Manufacturing defect	1	None	10	40	None
Ejection Charge	Too little charge	Not enough force to separate the body tubes.	7	Manufacturing defect	1	None	10	70	None
Tracking Smoke	Failed/improper ignition, incomplete combustion	Difficult to locate the rocket	1	Poor mixing from manufacturer or Improper care of charge (i.e. proper storage, no moisture)	2	None	10	20	None
Ejection Cap	Cracks/Damage	Ejection charge could fall out	9	Improper material selection or excessive stress or manufacturing defect	1	Visual Inspection	2	18	Handle rocket with care; regularly check integrity of cap
Ejection Cap	Too hard/brittle	Ejection charge would not properly blast upward	8	Improper material selection or excessive stress or manufacturing defect	1	None	10	80	Check integrity and replace if needed
Ejection Cap	Falls off prior to ignition	Ejection charge falls out or does not light properly	10	Improper seating	2	Visual Inspection	1	20	Ensure ejection cap is properly attached
Motor Retainer	Tearout	Motor could fall out of the body tube or goes up into the rocket	10	Improper material selection or excessive stress/fatigue or manufacturing defect or stripped threads	2	Visual Inspection	5	100	Run detailed simulations
Motor Retainer	Motor is not properly held in	Motor could fall out of the body tube or goes up into the rocket	10	Improper seating	2	Visual Inspection	2	40	Be cautious when installing motor
Motor Retainer	Cracks/Damage	Motor could fall out of the body tube or goes up into the rocket	10	Improper material selection or excessive stress or manufacturing defect	2	Visual Inspection	2	40	Handle rocket with care; regularly check integrity of retainer
Motor Retainer	Retainer threads strip	Motor not seated correctly & can cause instability	8	Excessive stress or manufacturing defect	1	Visual Inspection	1	8	Check integrity and replace if needed
Centering Rings	Glue doesn't bond correctly to airframe	Motor misalignment can cause instability	8	Old/Incorrect glue or assembly defect	3	Visual Inspection	4	96	Check integrity and redo if needed
Centering Rings	Not centered	Motor misalignment can cause instability	8	Assembly/Manufacturing defect	3	Visual Inspection	1	24	Be cautious when installing rings
Fins	Fluttering/Flapping	Rocket is unstable in flight	5	Glue doesn't bond correctly to airframe or opening is too wide or fins are too thin	4	Simulations and visual inspection	3	60	Check integrity and replace if needed; be cautious when gluing
Fins	Fins fall out	Rocket is unstable in flight	7	Glue doesn't bond correctly to airframe or opening is too wide or not properly attached	3	None	10	210	Check integrity and replace if needed; be cautious when gluing
Fins	Fins are cracked or damaged	Rocket is unstable in flight	5	Fins are dropped	2	Visual Inspection	2	20	Handle rocket with care

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Fins	Fins are misaligned	Rocket is unstable in flight	5	Uneven slots in airframe	4	Visual Inspection	1	20	Check fins regularly after transport
Fin Can	Fluttering/Flapping, unstable flight	Rocket is unstable in flight	5	Glue doesn't bond correctly to motor casing	4	None	10	200	Check integrity and replace if needed; be cautious when gluing
Shock Cord	Cord snaps	Parachute would be disconnected from rocket	10	Manufacturing defect or cord experiences too much force on separation	2	None	10	200	Run detailed simulations
Shock Cord	Cord becomes untied from Eyebolts	Parachute would be disconnected from rocket	10	Assembly defect	4	Visual Inspection	1	40	Be cautious when attaching cord; replace if needed
Shock Cord	Cord is tangled	Parachute may get caught and not deploy	8	Assembly defect	5	Visual Inspection	5	200	Be cautious when assembling rocket for a launch
Shock Cord Eyebolts	Cracked or broken	Shock cord detaches from rocket	10	Manufacturing defect	1	Visual Inspection	1	10	Check integrity and replace if needed
Quicklinks	Cracked or broken	Shock cord detaches from rocket	10	Manufacturing defect	1	Visual Inspection	1	10	Check integrity and replace if needed
Quicklinks	Unhooks from Eyebolds	Shock cord detaches from rocket	10	Assembly defect	2	None	10	200	Be cautious when assembling rocket for a launch
Parachute Canopy	Rips/tears preflight	Rocket does not slow down enough	7	Improper handling, manufacturing defect	1	Visual Inspection	1	7	Check integrity and replace if needed
Parachute Canopy	Rips/tears inflight	Rocket does not slow down enough	7	Improper deployment, getting caught on something, manufacturing defect	2	None	10	140	Be cautious when assembling rocket for a launch
Parachute Canopy	Improper deployment	Uncontrolled descent	10	Improper packing	3	Visual Inspection	3	90	Be cautious when assembling rocket for a launch
Parachute Cords	Tangled	Uncontrolled descent	10	Improper packing, getting caught on something	3	Visual Inspection	4	120	Be cautious when assembling rocket for a launch
Parachute Cords	Snapped	Rocket does not slow down enough	6	Manufacturing defect, deployment at excessive speed	2	None	10	120	Run detailed simulations
Drogue Canopy	Rips/tears preflight	Rocket does not slow down to adequate parachute deployment speed	7	Improper handling, manufacturing defect	1	Visual Inspection	1	7	Check integrity and replace if needed
Drogue Canopy	Rips/tears inflight	Rocket does not slow down to adequate parachute deployment speed	7	Improper deployment, getting caught on something, manufacturing defect, canopy gets burned	2	None	10	140	Be cautious when assembling rocket for a launch
Drogue Canopy	Improper deployment	Rocket does not slow down to adequate parachute deployment speed	6	Improper packing	3	Visual Inspection	3	54	Be cautious when assembling rocket for a launch
Drogue Cords	Tangled	Rocket does not slow down to adequate parachute deployment speed	6	Improper packing, getting caught on something	3	Visual Inspection	4	72	Be cautious when assembling rocket for a launch

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Drogue Cords	Snapped	Rocket does not slow down to adequate parachute deployment speed	8	Manufacturing defect, deployment at excessive speed	2	None	10	160	Run detailed simulations
Parachute Blanket	Ripped/torn	Flashes of heat could reach components and set them on fire	9	Manufacturing defect, improper packing	1	Visual Inspection	1	9	Check integrity and replace if needed
Heat Wadding	Amount is insufficient	Flashes of heat could reach components and set them on fire	9	Assembly defect, poor analysis	1	Simulations and visual inspection	3	27	Run detailed simulations
Heat Wadding	Wadding does not stop the heat	Flashes of heat could reach components and set them on fire	9	Manufacturing defect	1	Simulations and visual inspection	4	36	Run detailed simulations
Nose Cone	Disconnects from parachute	Nose cone becomes a projectile	10	Manufacturing defect, assembly defect	2	Visual Inspection	5	100	Check integrity of parachute attachment; redo if needed
Nose Cone	Falls off prior to ejection	Rocket loses aerodynamic efficiency and does not go high enough	6	Assembly defect; poor design	3	Visual Inspection	2	36	Ensure fit is good during assembly
Nose Cone	Threads become loose	Payload falls out	6	Manufacturing defect, assembly defect	3	Visual Inspection	2	36	Ensure fit is good during assembly
Nose Cone	Does not release when needed	Nothing gets deployed, no parachute	6	Too Big	3	Visual Inspection	2	36	Ensure fit is good during assembly
Bulkhead	Bulkhead falls out	Payload falls out	6	Epoxy doesn't bond correctly	1	None	10	60	Check integrity and replace if needed; be cautious when gluing
Coupler	Decouples inflight	Rocket becomes two pieces	9	Too small	2	Visual Inspection	3	54	Ensure fit is good during assembly
Coupler	Does not release when needed	Nothing gets deployed, no parachute	10	Too big	3	Visual Inspection	3	90	Ensure fit is good during assembly
Main Rocket Body	Body buckles inflight	Rocket falls apart, yields	10	Manufacturing defect	1	Simulations and visual inspection	1	10	Run detailed simulations
Launch Rail Guides	Rails separate from body	Rocket launches at the wrong angle	8	Screws or epoxy come loose, improper seating, rips through body	1	None	10	80	Check integrity and replace if needed; be cautious when gluing
Payload Release Mechanism	Does not release when needed	Payloads stay within the rocket, payloads launch at too low of an altitude	6	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	36	Run electronics tests to ensure functionality; replace if needed
Payload Release Mechanism	Releases too early	Rocket interior and chutes gets toasted with a pyro charge	9	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	54	Run electronics tests to ensure functionality; replace if needed
Drogue Ejection Mechanism	Does not release when needed	Drogue stays within the rocket, drogue deploys at too low of an altitude	7	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	42	Run electronics tests to ensure functionality; replace if needed
Drogue Ejection Mechanism	Releases too early	Rocket interior and chutes gets toasted with a pyro charge	9	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	54	Run electronics tests to ensure functionality; replace if needed
Parachute Ejection Mechanism	Does not release when needed	Parachute stays within the rocket, parachute deploys at too low of an altitude	9	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	54	Run electronics tests to ensure functionality; replace if needed

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Parachute Ejection Mechanism	Releases too early	Rocket interior and chutes gets toasted with a pyro charge	9	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	54	Run electronics tests to ensure functionality; replace if needed
In-Flight Payload Deployment	Early deployment of parachute	Rocket is pulled into an incorrect trajectory, deployment shell detaches from rocket	7	Improper packing, assembly defect, electronics timing defect	3	Simulations and visual inspection	2	42	Run electronics tests to ensure functionality; replace if needed, Ensure fit is good during assembly
In-Flight Payload Deployment	Servo motor does not activate	Payload stays within the rocket	5	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	5	75	Run electronics tests to ensure functionality; replace if needed, Ensure fit is good during assembly
In-Flight Payload Deployment	Black powder charge does not fire	Payload stays within the rocket	5	Manufacturing defect, Electrical defect, Assembly defect	4	Simulations and visual inspection	2	40	Perform a test fire of the powder charge, Store powder properly before launch
In-Flight Payload Deployment	Parachute does not deploy	Payload stays within the rocket	5	Packing defect	4	Visual Inspection	2	40	Be cautious when assembling rocket for a launch
In-Flight Payload Deployment	Beeper does not activate	Payload is lost	4	Manufacturing defect, Electrical defect	3	Electronics Tests	2	24	Run electronics tests to ensure functionality; replace if needed
Ground Ejection Mechanism (GEM)	Does not release when needed	Payload stays within the rocket	6	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	36	Run electronics tests to ensure functionality; replace if needed
Ground Ejection Mechanism (GEM)	Releases too early	Payload would eject before landing	5	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	30	Run electronics tests to ensure functionality; replace if needed
GEM Bearings	Bearings pop off their structure	Payload does not orient correctly for deployment	5	Manufacturing defect, assembly defect, poor tolerancing, physical damage	1	Visual Inspection	1	5	Run detailed simulations
GEM Structure	Structure becomes jammed	Payload does not orient correctly for deployment	5	Internals interact with center axle, physical damage	4	Visual Inspection	5	100	Run detailed simulations
GEM Structure	Breaks	Payload does not orient correctly for deployment	5	Manufacturing defect, assembly defect, poor tolerancing, physical damage	2	None	10	100	Run detailed simulations
GEM Stability System	Weights dislodge from structure	Payload does not orient correctly for deployment	5	Manufacturing defect, assembly defect, poor tolerancing, physical damage	2	Visual Inspection	5	50	Run detailed simulations
GEM Hatch	Hatch breaks off body tube	Payload is lost	6	Poor axle design	2	Lab Testing	3	36	Test load bearing capability
GEM Axle	Breaks along length	Rocket motor separates from parachute	10	Improper design	3	Lab Testing	3	90	Test load bearing capability
GEM Axle	Threads tearout	Rocket motor separates from parachute	10	Improper design	4	Lab Testing	3	120	Test load bearing capability

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GEM Couplers	Threads tearout	Rocket motor separates from parachute	10	Improper design	4	Lab Testing	3	120	Test load bearing capability
GEM Couplers	Couplers detach from rocket	Rocket motor separates from parachute	10	Assembly defect	2	Lab Testing	3	60	Test load bearing capability
GEM Hatch	Outside debris prevent door from opening	Payload does not deploy	3	External blockages	5	None	10	150	Attempt to aim rocket into wind upon launch
GEM Stability System	Outside debris prevents tube from orienting	Payload does not deploy with proper orientation	3	External blockages	5	None	10	150	Attempt to aim rocket into wind upon launch
Nose Cone Payload	Hatch door gets stuck	Parachute is unable to deploy	10	Too tight of tolerance for door to release	2	3D prototype and testing	3	60	Print iterations of the door until loose fit tolerance.
Nose Cone Payload	Hatch door gets stuck	Parachute is unable to deploy	10	Electronics malfunction (battery dies, code incorrect, etc) and servo does not initiate or servo is damaged in flight	3	Electronic test	3	90	Run electronics tests to ensure functionality; replace if needed
Nose Cone Payload Parachute	Parachute does not un-foil itself	Fast descent speed	8	Shock cord potentially gets parachute stuck or rips	2	Have a good folding technique to place parachute in hatch	2	32	Try to perform a smaller test on a tall garage to see if it can unfold
Nose Cone Payload Shock cord	Shockcord rips out of nose cone	Fast descent speed and loss of parachute	8	Shock cord is not tied down correctly or holding structure breaks	1	3D prototyping and lab testing	2	16	Make sure the shock cord is fully tight and make preventative measures like taping knots, etc.
Nose Cone Payload Shoulder	Too tight of a fit	Nose cone cannot slide out of rocket during release	10	PLA too tight or not sanded down enough	5	Lab Testing	2	100	Bring extra sandpaper day of launch to make sure the shoulder is not too tight since PLA and cardboard is affected by weather
Nose Cone Payload electronics unit in shoulder	Thread tearout	The shoulder that houses electronics and payload will disconnect from main parachute	8	Improper design	2	Lab Testing	4	64	Make sure to give enough length of threads and tolerance correctly to make sure they stay in place
Nose Cone Payload payload unit in shoulder	Thread tearout	The shoulder that houses payload will disconnect from main parachute	8	Improper design	2	Lab Testing	4	64	Make sure to give enough length of threads and tolerance correctly to make sure they stay in place
Battery	Improper output voltage	damage to components, components do not turn on	10	traces not labeled or connected to proper terminals, choice of battery	7	use LED's as a peripheral to show if proper voltage is being supplied to the PCB	1	70	Use a new battery,
Battery	improper current supplied to system	no current sent to components, damage to components	10	Manufactured defect, traces not labeled or connected to proper terminals	3	calculating the current draw of peripherals and following datasheets	2	60	Use a new battery, change peripheral models

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MCU	Programing errors	desired commands are not initiated on the board and corrupted data transfer occurs	10	invalid trace length, not paying attention to orientation labels, high density IC, Orientation	6	Running code simulations	5	300	Test with development board, check wiring and chip, replace and resolder if necessary
MCU	Watchdog resets	No commands are initiated on the board and no data transfer occurs, coding errors,	10	overloading data with too much information due to high sampling rate on multiple sensors	2	selecting sampling rates by importance of information	1	20	changing the sampling rate if this persists
MCU	damaged	No commands are initiated on the board and no data transfer occurs	10	too much voltage supplied, physically damaged,	2	double checking Kicad labels for the pins and we can also use a silk screen to write the texts on the PCB to confirm after printing	1	20	Test with development board, check wiring and chip, replace and resolder if necessary
Regulator	Desired voltage isn't reached	supplying too much power or not enough power to components leading to either overheating/damage	10	amount of supplied voltage, model of regulator, compatibility to chosen MCU	9	use LED's as a peripheral to show if proper voltage is being supplied to the PCB	1	90	Redesign regulator if original is wrong design; replace components to double-check
Regulator	Improper output current	can't support current draw of peripherals	10	not choosing correct models for regulator for desired current output	2	researching models that support current requirements of peripherals	1	20	Redesign regulator if original is wrong design; replace components to double-check
IMU	Doesn't measure or transmit flight data to MCU	no flight data to be sent to ground station	7	improper connections, wire improperly built PCB	5	use LED's as a peripheral to show if proper voltage is being supplied to the PCB Test PCB on breadboard for measurements, label pins using silkscreen.	6	210	Test with development board, check wiring and chip, replace and resolder if necessary
IMU	inaccurate readings	inaccurate readings sent to ground station	1	acceleration of rocket, low transmission rate	7	using models that can withstand high speeds and have high accuracy	4	28	replace IMU model
Altimeter	Doesn't measure or transmit flight data to MCU	actuators never activated, payloads never get released from rocket, no flight data sent to ground station	10	improper connections, wire improperly built PCB	5	Use LED to test voltage through device. Test PCB on breadboard for measurements, label pins using silkscreen	1	50	Test with development board, check wiring and chip, replace and resolder if necessary
Altimeter	inaccurate readings	inaccurate readings sent to ground station, payloads/parachutes don't release at desired altitudes	10	acceleration of rocket, low transmission rate, rapid pressure change, out of sensing range	5	using models that can withstand high speeds and have high accuracy	1	50	replace altimeter
Transceiver	does not convert data from MCU into analog	no data received on ground station, rocket can not be recovered, data corrupted	7	having the transceiver too close to other components, model of transceiver used, wiring and connections	3	Use a conversion component onto board to transfer to google maps	1	21	Test

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Transceiver	does not send data to ground station	no data received on ground station, rocket can not be recovered, data corrupted	7	antenna isn't supplying desired gain, groundstation out of range model of transceiver	8	use an active antenna to increase gain and cover wider distance	1	56	Test
GPS	Failure to detect satellites,	Low tracking accuracy or failure to track,	8	model used, signal strength	3	using a module that satisfies requirements	4	96	Test
GPS	GPS locks during ascent	current location of rocket is not sent to ground station only position before lock,	7	model used, acceleration of rocket	1	choosing a module that supports high speeds	1	7	replace module
Actuator	Desired motion not reached	springs aren't released when desired altitude reached, door doesn't open to release payloads from body tube	10	not enough voltage supplied to power actuator, draws too much current from the battery/regulator	10	Measure voltage and current when testing actuator movement	2	200	Replace actuator
Beeper/Buzzer	Doesn't go off	Rocket/Payloads harder to find, possibly never recovered	8	no power supplied to beeper, damaged component	2	proper wiring	2	32	Test
Beeper/Buzzer	Distance the sound travels	Rocket/Payloads harder to find, possibly never recovered	8	speakers covered/blocked by environment	7	Model used for frequency and range	1	56	Noise range test
Antenna	doesn't transmit at a frequency that covers 5k+ feet	data wont get picked up by ground station, difficult to recover rocket, cant read flat data	10	bent antenna, too close to other components leading to noise and corruption, disconnecting from PCB	8	Use commonly used models,	1	80	Range Test
Surface Mount Device Components	Doesn't fit traces on PCB	not being able to see components, can't solder components, can't properly control current flowing through system	10	using components (capacitor, resistor, inductor) smaller than an 805, ordering the wrong parts	2	Measure each component, correct size during PCB design	1	20	Redesign PCB to accommodate correct size
Male/Female headers	Falls out during flight	peripherals aren't powered by main power supply, data isn't transferred to the MCU from peripherals	10	not properly soldered, vibration, improper orientation on PCB, Improper pin labels on Kicad	2	reviewing PCB labels and checking names and connections and printing with a silkscreen when soldering connections	4	80	Check wiring, replace and resolder if necessary
Male/Female headers	not enough surface area on board to support all headers	not being able to house necessary peripherals	3	improper wire connections, improperly built PCB	4	using wires to connect peripherals if necessary	1	12	Check size on PCB and headers to see if they match
Wire Connection	no voltage and current transferred to peripherals	no power delivered to components and no functions occur	10	Connections fall out during flight, wire breaks, improper wiring	2	keeping boards attached using male and female headers to avoid using wires	1	20	Check connections, measure voltage and current
SD card	Failure to store data	not being able to review all/any flight data	5	lack of storage space, data never sent due to improper wiring	3	labeling wires and choosing SD card with ample storage	10	150	check wiring, leave labels, choose different SD card
Electronic Bay	Moisture leaks	Electronics could short	10	Cracks or leaks in the rocket, exposing the electronic bay	2	Using airtight structures	7	140	Check holes and use sealant

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Electronic Bay	Bad mounting	Components are loosen and can damage other parts	10	Screws not secured to board, bay not contained to rocket	2	Using all corners on PCB for screw holes	3	60	Using long screws and lock nuts
Overall Electronics system	Overheating	all components stop functioning	10	too much voltage supplied, poor regulation, improper connections, lack of insulation	2	checking connections, using regulators that can supply desired voltage, proper spacing	5	100	Test
USB	Doesn't match port on device	data isn't sent to the terminal to display flight data	6	not choosing proper component when designing ground station PCB	1	Using a MCU module that supports a USB-A connection	1	6	Test
External Logging Device	Doesn't properly show flight data	terminal does not display the data and location is not displayed on google maps	7	Ground-station software cannot decode telemetry packets due to packet loss, baud mismatch, or improper data formatting	2	Running code simulations to make sure data is being converted from the MCU to terminal before launch	1	14	Test
External Logging Device	no battery	terminal does not display the data and location is not displayed on google maps	7	not charging laptop or phone	1	making sure everyone's devices are charged	1	7	charging before demo
External Logging Device	no device present	terminal does not display the data and location is not displayed on google maps	7	no one brings a device that will display flight data	1	having at least 2 people confirm that they have their devices present	1	7	bring a device
PCB's	manufacturer does not properly sketch wire tracings	power isn't distributed properly, boards don't perform intended goals	10	issues at the shop, wire tracings too thin and missed	2	having a set trace length for wires, avoiding small components	5	100	manually sketch wires in SD2 lab
PCB's	stencil doesnt match pcb outline	difficulty soldering parts, components not aligned system fails	2	not adding the proper measurements when ordering PCBs for stencil lengths, not ordering the stencils at all	3	confirming the dimensions of the stencils when making orders	10	60	solder without the stencil and be careful
PCB's	burn through wire connection while soldering	no information or power sent to a/multiple components, intend results not met	7	using too much heat when soldering, holding the pen to the PCB for too long	1	following SD recommendations for soldering temperature and being safe	1	7	solder on new wire to recreate the connection
PCB's	not having enough space for peripherals	not having all peripherals integrated onto main PCB using male/female headers	7	not taking measurements of peripherals into account when adding the headers onto the main PCB	3	using the measure tool on the silk screen to outline peripheral dimensions	2	42	stacking headers or having peripherals at an angle

A Failure Modes and Effects Analysis (FMEA) was conducted to identify the critical areas of concern for the selected concepts. For each design component, failure modes were identified to represent the various causes of a part's inability to perform its required function. The team listed potential causes for each failure mode to serve as a base for determining ways to mitigate the risk it presents. To determine which failure modes were most significant, the

potential effects of each were analyzed and given a severity (Sev) score out of ten. Additionally, the probability (Prob) and determinability (Det) of each failure mode were similarly ranked.

When evaluating severity, a basic framework was laid out to guide the team's ratings: a score of 1 represented negligible damage or minor costs as a result of the failure mode; a score of 5 represented the potential for medium damage or a minor risk of injury; and a score of 10 represented the possibility of a major injury, major cost, or a fatality. Additionally, from the context of this project as part of a capstone class, failure modes that might result in critical requirements not being met (such as the safe deployment of payloads) were assigned severity scores of up to 7. However, severity scores of 8 through 10 were reserved for effects that lead to significant bodily or property harm outside of the scope of the project to ensure that safety is always paramount when making engineering decisions.

The probability score was determined on an approximately linear scale. A score of 10 roughly represents a 10% chance of the failure mode occurring, and a score of 1 represents a case that almost never occurs and is a negligible concern. Probability scores of 1 were assigned very sparingly, typically for failure modes caused by manufacturing defects of simple parts. The determinability score shows how likely the team is to uncover the causes of the failure and correct them before the failure occurs. Higher detectivity scores are, counterintuitively, less detectable failure modes. A 10 in this category is a failure mode that is impossible to detect before it fails, and a 1 is a failure mode that can be detected very easily and with minimal testing.

All of the severity, probability, and detectivity scores were combined into a Risk Priority Number to compare the significance of each failure mode. Risk Priority Numbers (RPN) were calculated by simply multiplying the Severity, Probability, and Detectivity.

$$RPN = Sev * Prob * Det \quad (1)$$

A significant example is the motor's ejection charge not firing properly (RPN of 200), leading to the parachute not deploying properly and possibly an uncontrolled descent. An uncontrolled descent is automatically assigned a severity score of 10, as it can cause significant property damage or death. Additionally, the detectivity score was deemed a 10, as the cause of this failure mode is poor-quality grains, something that will not be discoverable without advanced testing and destruction of said grains. However, the probability of this occurring is low

so long as the team properly stores the motor and its components. Thus, probability was given a 2, and the failure mode was deemed significant but preventable.

Another example is the main parachute chords getting tangled (RPN of 120). If the parachute fails to inflate, the rocket can again go into an uncontrolled descent, and so the severity is, again, a 10. This time, however, the failure mode is detectable by checking the parachute packing before launch, leading to a detectability score of 4 to account for human error. The probability is also low and was given a 3, leading this failure mode to be less significant overall than the ejection charge as long as the group performs its preflight checks properly.

On the ECE side, one failure mode is the regulator's voltage and current function, namely that the desired output voltage or current is not reached. This could be due to the regulator model, the input voltage, or the input current. The potential effects of the failure mode could cause sensors to overheat or fail to function. These failures have high severity (10), a high probability of the voltage failure mode (9), and a low probability of the current failure mode (2). It should be very easy to detect, as the peripherals would not be functioning. The regulators have LEDs to indicate the supply voltage.

This highlights the purpose behind an FMEA. By this method, critical sources of risk can be dissected and thoroughly accounted for to give the project its highest chance of success. By comparing each of the modes and their RPN, the team can determine which risks need to be specially accounted for in the design stage, with redundancy if necessary, and which risks can be mitigated by basic checks.

Modeling and Analysis

The system concept is considered complete and well-defined because the team has clearly identified the main objectives, functional necessities, and subsystem interactions needed to achieve a successful flight. Each major component including the airframe, recovery systems, avionics, and payload deployment mechanisms has been mapped to the specific customer requirements. Complete designs and schematics have been documented, and the interfaces between each subsystem have been reviewed to ensure that the subsystems and electronics integrate properly and operate feasibly within the constraints of the rocket.

To visualize the system, the team has developed detailed CAD models of each individual subsystem and the overall rocket components. Below will show more in-depth of each subsystem - nosecone payload, in-flight payload, and ground payload. Additionally, there are electrical schematics of the PCB design and individual electrical components (i.e. ground station, voltage regulator, altimeter, inertial measurement unit, GPS, buzzer, and microSD card). The electrical schematics showcase the avionics architecture such as how the boards are connected, power distribution, and servo actuation for payload deployment. All of these schematics and CAD models help visualize the complete system and provide a high-level system understanding to verify the integration of each subsystem.

The modeling and analytical approach is an accurate representation of the physical rocket system because it combines preliminary engineering analysis, validated simulation tools, and planned electronic and hardware testing. The team initially used OpenRocket to model the full vehicle breakdown, which incorporates simulations that can show velocity, acceleration, drag, thrust, altitude, and stability based on established underlying mathematical theory.

After flight modeling and simulation, detailed CAD assemblies were developed for each subsystem to verify the fit, mass, and integration between each component. Critical components such as the recovery system were evaluated through simulations, and the servo functions will be evaluated through electronic testing. Where there is analytical uncertainty present, the team will perform lab testing to validate the assumptions and refine the whole model. Because this approach contains simulations, physical models, and empirical validation, it is expected to closely represent real flight behavior and provide reliable predictions of the flight performance.

There are many boundary conditions and constraints on the overall design. The rocket is subject to boundary conditions and engineering constraints that govern the performance and integration of the subsystems. There are also geometric and packaging constraints for each payload deployment mechanism and electronics systems. These designs are fixed by the inner diameter of the body tube (5.38 inches), nose cone volume, and coupler lengths.

There are also mass and stability constraints to the rocket that are imperative for a successful flight. The rocket stability must be within ~ 1.5 cal to ~ 2 cal to maintain a stable flight. The mass is also constrained to the size of the motor class. The team ran into this when the rocket was heavier than anticipated and required a class increase in motor to maintain the necessary altitude.

There are also electrical constraints with power that limit the avionics and subsystem sections. The battery capacity, voltage regulator, and current must be able to flow through the system to support any electronic systems without causing signal instability. The orientation of Radio frequency components are also a constraint for the telemetry system as too much electrical noise can cause signals to be corrupted so we must be careful when placing components.

Manufacturing constraints is a big topic as a bulk of the team's components are 3D printed components. This requires the nose cone, shoulder, in-flight descent payload, and ground payload to maintain the needed sufficient structural integrity.

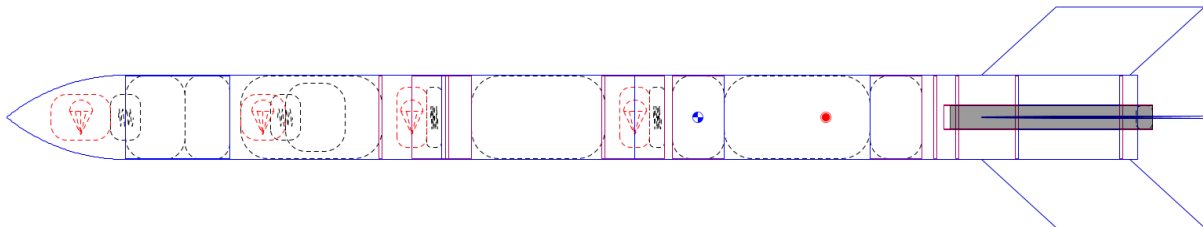
Finally, a critical operational constraint involves the reliable sequencing of the payloads, drogue, and main parachute events, and making sure everything is compliant with NAR safety standards. Together, these boundary conditions define the allocated space and how the team must model and analyze each subsystem to ensure the final system is both physically realistic and will be successful.

Overall Rocket

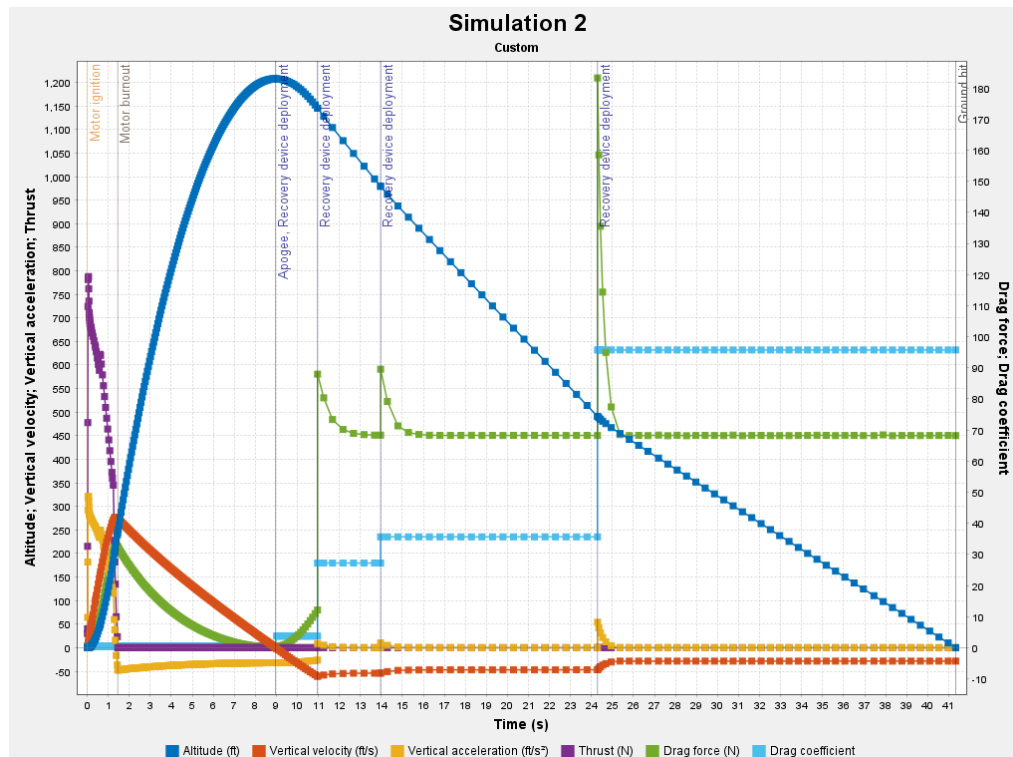
The overall rocket design establishes the structural foundation and aerodynamic profile required to meet the target apogee and maintain the needed stability margin of 1.5 cal or higher (not exceeding 2 cal). The airframe distribution, fin geometry, and general mass properties are used to maintain said stability, reach needed altitude, and simulate dynamic properties. Having

Flight configuration: [J500-0]
Apogee: 1208 ft
Max. velocity: 277 ft/s (Mach 0.248)
Max. acceleration: 321 ft/s²

Rocket Fight Simulation



Side View of Rocket



Rocket Simulation with J500 Motor

The rocket is designed to best accommodate each subsystem that will be deployed during the final flight. In the side view of the rocket, the nose cone contains a 9-inch tip where the parachute is located in the forward section. The shoulder includes two mass blocks:

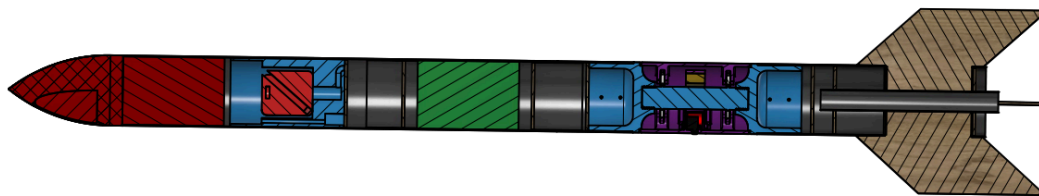
approximately 5 inches allocated for the electronics and approximately 3 inches at the bottom end for the cargo payload.

Below the nose cone section, there is a 9-inch mass component that represents the in-flight descent payload. The 4 in $\times$ 4.5 in CubeSat payload is positioned about 2.5 inches above the shell. This 2.5-inch space houses the servo motor and the black powder ejection system.

The first coupler contains the drogue parachute, which slows the rocket prior to main parachute deployment. The central mass component houses the electronics bay, which contains the main power supply. Beneath the electronics bay is a second coupler that separates for main parachute deployment.

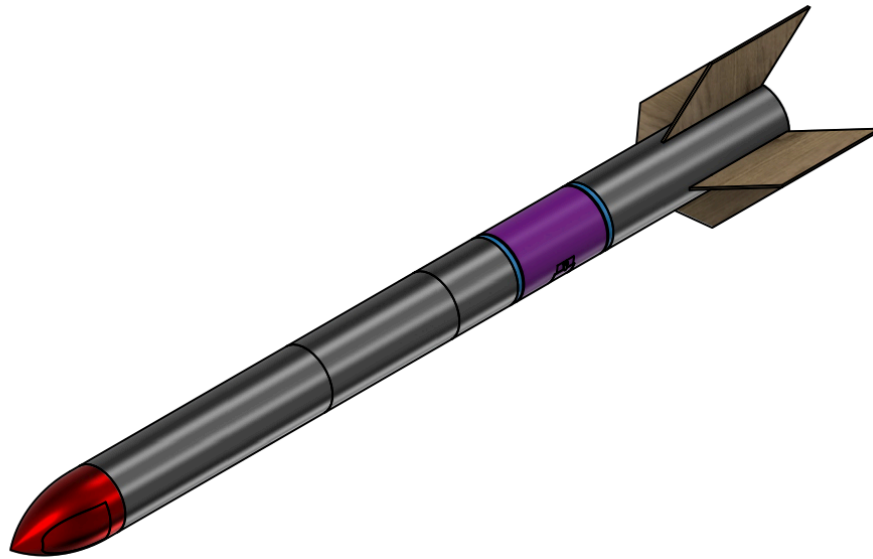
The bottom portion of the rocket represents the rotating body tube. The two end pieces in this section are internal 3D-printed couplers that secure the rocket components together with the rotating body tube. This section is responsible for deploying the rover payload after the rocket lands.

Originally, the team assumed the rocket mass would be lower and planned to use an I500 or I600 motor based on early simulations. However, due to the significant weight increase from the predominantly plastic structures, the team re-evaluated the propulsion requirements and conducted updated simulations. Based on these results, the team selected the J500 motor as a suitable alternative, which is expected to bring the rocket to approximately 1,200 ft.



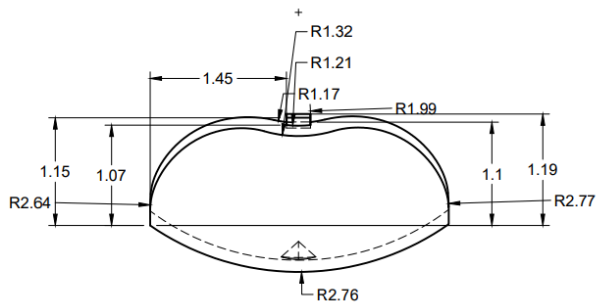
SECTION A-A
SCALE 1:7

Section View of Rocket Assembly

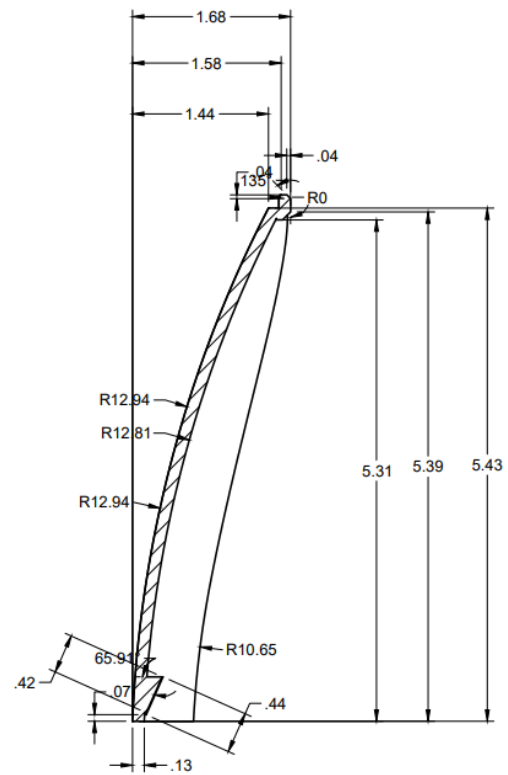


Isometric View of Rocket Assembly

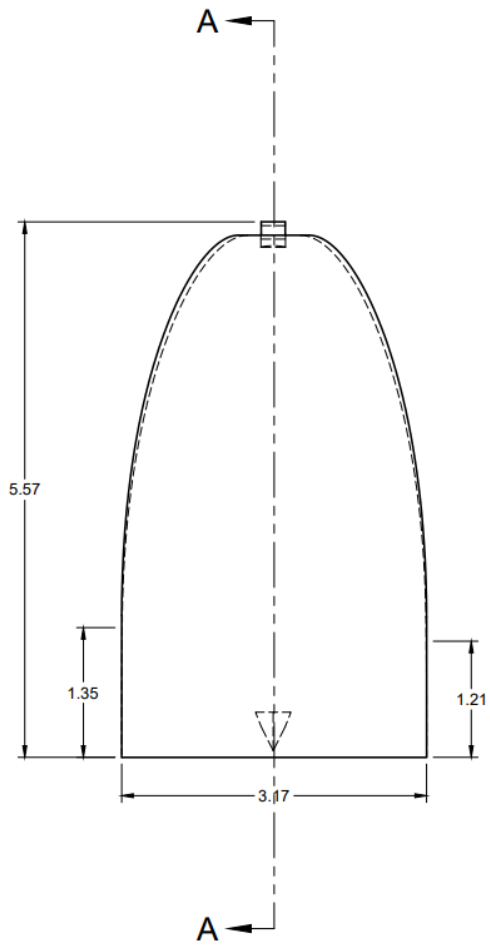
Nose Cone Payload



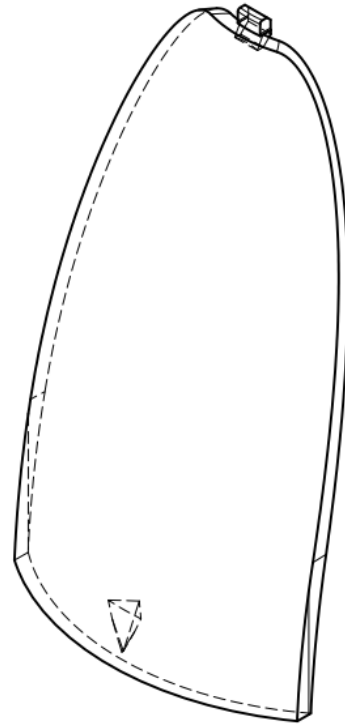
Top View of Hatch Door



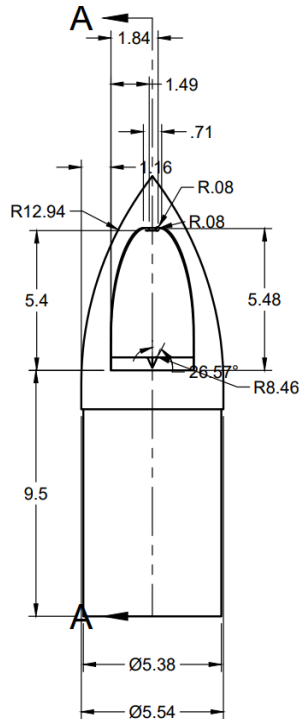
Section View of Hatch Door



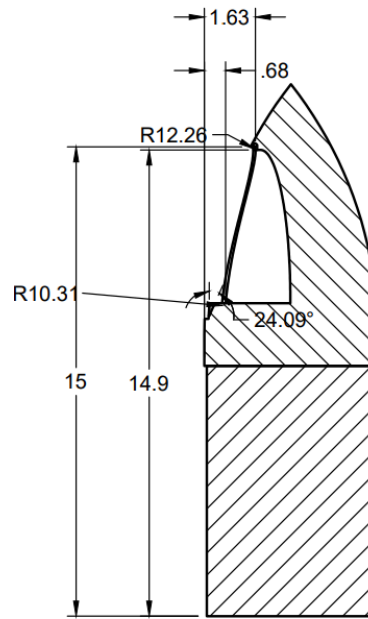
Front View of Hatch Door



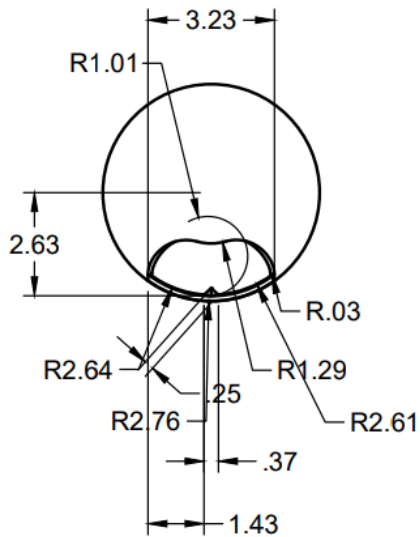
Isometric View of Hatch Door



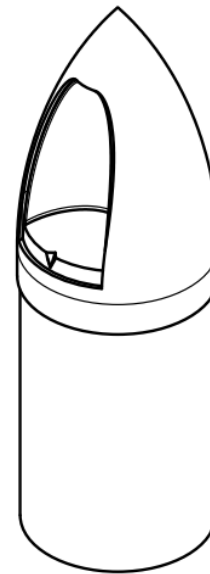
Front View of Nose Cone



Section View of Nose Cone



Top View of Nose Cone



Isometric View of Nose Cone w/ Shoulder

The nose cone payload subsystem is designed to operate within tight volumetric and size constraints while maintaining the necessary mass distribution at the forward part of the rocket (~2.05 lbs). The purpose of this nose cone is to fully detach from the vehicle in order to create an opening in the rocket for the in-flight descent payload to exit. The design includes a curved

cut-out bay where the parachute will fold and sit. A hatch door closes the bay and is held in place with a servo motor. The servo motor will initiate once apogee is detected through an altimeter, allowing the door to open and the parachute to naturally fall out. The hatch door remains attached to the nose cone so that it stays with the full unit after deployment.

The electronics and payload cargo will be housed in the shoulder of the nose cone. The height of the nose cone tip is 9 inches, and the shoulder section that includes the electronics and cargo package is 8 inches. The nose cone tip will be printed separately due to the height limitations of the 3D printer. The shoulder was therefore designed to thread into the bottom of the nose cone tip to securely hold both sections together. A similar threaded layout will be used underneath the electronics section to attach the payload cargo package below. This modular approach ensures that each component can be printed separately and properly integrated into the main rocket body tube.

The team selected an ogive nose cone shape due to its highly efficient aerodynamic characteristics and its ability to minimize drag at higher speeds. This geometry helps the rocket reach higher velocities and altitudes by reducing aerodynamic resistance. A complete CAD assembly of the nose cone will be developed to show all individual components, including the nose cone tip, hatch door, servo motor, electronics section, and payload sections.

After fabrication, the printed components will undergo lab testing so that any necessary modifications can be made quickly and efficiently. Each part will be weighed to ensure the actual masses remain consistent with the simulated values, since increases in nose cone weight can significantly affect the overall rocket stability. Once the electronics are fully integrated and connected, additional lab testing will be conducted to verify that the servo motor and deployment sequence function as intended.

Key parameters relevant to the nose cone design include air density, rocket velocity and max acceleration from OpenRocket, and dynamic pressure. Another critical parameter is the payload mass, which must remain close to the simulated value to preserve the rocket's stability characteristics. For the electrical system, important parameters include battery capacity and voltage, voltage regulator output, barometric accuracy of the altimeter, and sensor sampling rate. Any missing parameters will be identified and refined through laboratory testing of the

electronics and through fit-check testing of the shoulder within the rocket airframe to ensure the design is fully validated.

In-Flight Payload

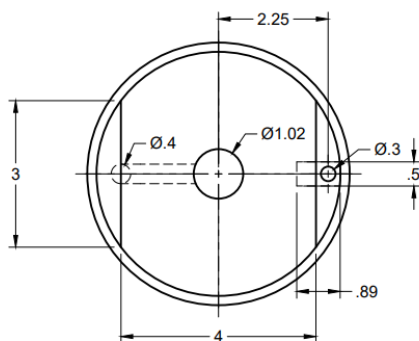
For the in-flight payload deployment, the team has created a shell which will be 3D-printed that holds the payload and its ejection charge. This shell will be located towards the top of the rocket, right below the nose cone. When designing this subsystem, the team considered the options on how to hold the payload before deployment, and how to open the rocket body so it could be ejected. The solution for this was to make the nose cone's deployment a passive system, where the deployment of a parachute would pull the nose cone off, leaving an opening for this payload deployment sequence. In order to ensure that the payload did not move before the team was ready to deploy it, a servo motor will be used to hold it inside the shell. This servo will have a small arm over top of the payload which will rotate open, then a black powder charge beneath the payload will eject it from the rocket. At this point, the parachute attached to the payload would deploy from the wind around it. The team's risk assessment was completely accounted for in the design and will ensure its efficacy.

The model for this deployment system was created in SolidWorks. By using CAD software, we can accurately model measurements and interactions, and also can directly produce the physical model by 3D-printing it once finished. By importing the model to ANSYS, we can also conduct FEA calculations to ensure an estimated structural stability. 3D-printed models are not always reliable during analysis due to the different levels of infill that are used when printing. FEA calculations generally consider a 100% infill and homogeneous material, however 3D printers work in layers. It is a good start, but actual testing once printed is also necessary. By using CAD software for our models, we can also ensure that the measurements fit with all other designed subsystems on the rocket, as well as ensure compatibility with the electronics systems, which will need to be installed in each payload as well as within the rocket body.

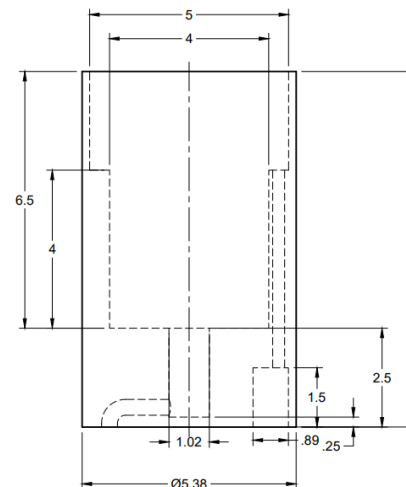
Below is the modeled in-flight payload deployment assembly. The outer shell will friction fit within the rocket body, and will sit in a centering ring that is flush with the drogue coupler. Within the bottom of the shell, there is a compartment for the black powder charge to reside and a hole for the electric match to be connected with the electronics below. The charge

will be composed of black powder within a finger of a latex glove, then the electric match will be placed inside and the hole will be taped shut. This will ensure that the powder stays together until ignited, and will blow through the glove and eject the payload. On the bottom right of the deployment shell, there is a housing for the servo motor. From the servo, a shaft will go up the right side and from there, the arm will be attached over top of the payload to hold it in place. Before the powder charge is detonated, this servo will spin to move the arm out of the way of the payload's path. In the middle of the shell is the payload itself simulating a typical cube-sat. As we are only simulating these payloads, the actual "cube-sat" will just be a box with a parachute attached at the top hole. Within the payload is a diagonal slot. This is where the electronics for an onboard audible beeper will be installed, and will allow us to locate the payload to recover it after landing. The space above the payload will be utilized for the parachute.

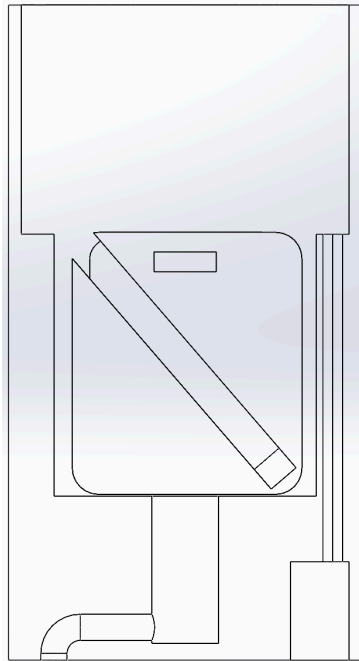
Below is the front view, top view, and isometric view of the payload shell that will be friction fit into the body tube.



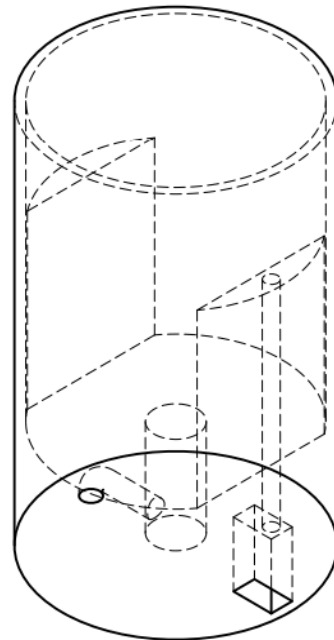
Top View of In-Flight Payload Shell



Front View of In-Flight Payload Shell

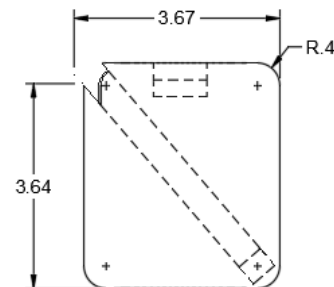
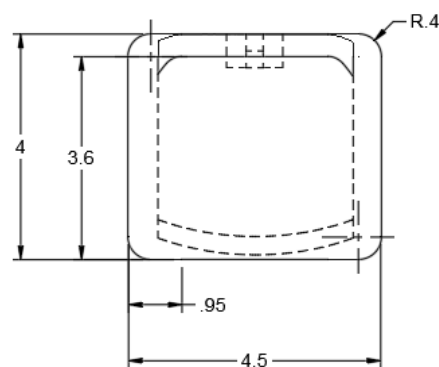
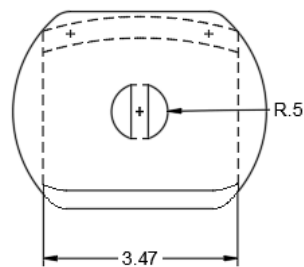


Side View with Payload



Isometric View of In-Flight Payload Shell

Below is the top view, side view, and front view of the “cubesat” payload that will be inserted into the payload shell above.

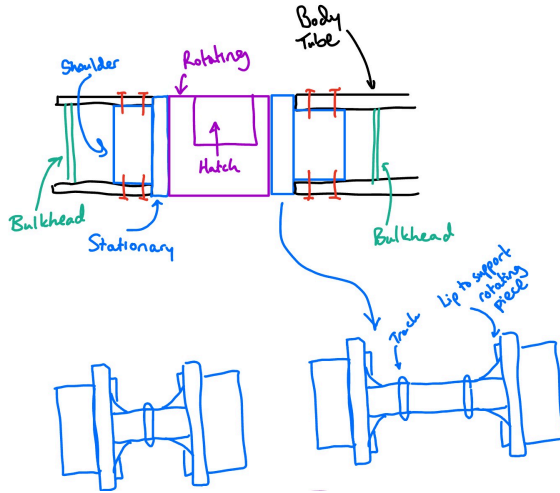


Ground Deployment

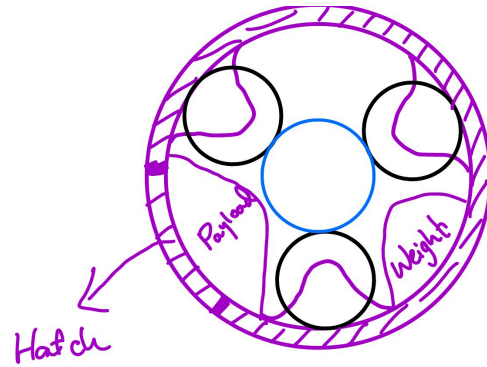
When deploying the ground payload (simulated rover), the system has two critical functionalities: first, that the payload must successfully separate itself from the rocket; and second, that the payload must be oriented right side up when deployed. As the payload is symbolic of a rover deployment, the system needs to be designed to support the rover's operation.

The original design concept (as detailed in previous milestones) was to use an internal rotating mechanism that used passive stability to orient the rover properly along the centerline of the rocket's body tube. From there, a spring would push the car out an open end of the body tube. This design needed to be reworked, however, because no solution could be found to separate the rocket body tube to create said opening without A) blocking off the ejection mechanism from a pyro charge with a bulkhead and rendering it useless, or B) damaging the ejection mechanism with an unblocked pyro charge.

A new system was designed that flipped the original idea. Instead of a fixed external structure supporting a rotating internal orientation system, the rocket would utilize a rotating external orientation system supported by a fixed internal structure. This allows bulkheads to be used when separating sections of the rocket and properly protects the deployment mechanism without blocking it. The concept is to mount an internal cylinder in between two sections of regular body tube which will serve as the axle around which the external orientation system will rotate. The external orientation system will mimic the size of the body tube to reduce drag while rotating on bearing wheels that roll on the internal axle. Weights will be added to one side of the external system to create a stable point that the system will naturally return to regardless of the rocket's orientation, thus ensuring the rover will be properly oriented. Initial sketches of the concept are included below. Objects in purple rotate, while objects in blue are fixed.



Preliminary Rotating Tube Design



Preliminary Section View of Design

The center cylinder would be responsible for holding two halves of the rocket together, and would need to be strong enough to withstand all in-flight loading conditions. Using principles of solid mechanics, hand calculations were performed for the two primary stresses the axle will experience during a launch: compressive stress and bending moment. As the cross sectional area is a circle, the following formulas were defined.

$$\text{Compressive Stress: } \sigma = \frac{F}{A}$$

$$A = \frac{\pi}{4} * d^2$$

$$\frac{\pi}{4} * d^2 = \frac{F}{\sigma}$$

$$d_{min} = \sqrt{\frac{4}{\pi} * \left(\frac{FOS * F}{\sigma_{max}} \right)} \quad (1)$$

$$\text{Bending: } \sigma = \frac{Mr}{I}$$

$$I = \frac{\pi d^4}{64}$$

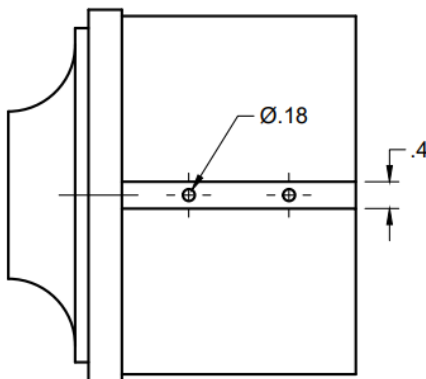
$$\sigma = \frac{64 * M * r}{\pi d^4}$$

$$\sigma = \frac{4M}{\pi r^3}$$

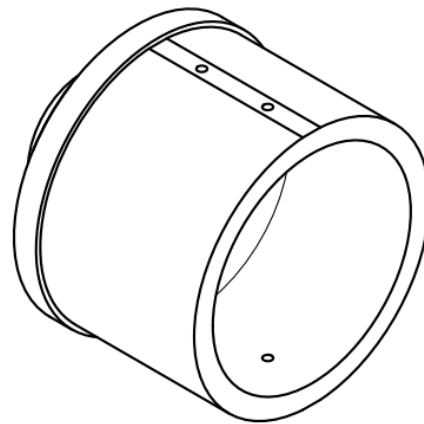
$$d_{min} = 2 * \sqrt[3]{\frac{4 * FOS * M}{\pi \sigma_{max}}} \quad (2)$$

The yield strength of PLA (the chosen material for this section) can range from 25 MPa to 50 MPa. To ensure the integrity of the rocket, 25 MPa was used in calculations. This figure assumes a 100% infill of PLA. Due to weight concerns, 100% infill is not the team's first choice. Thus, the maximum bending moment was intentionally over estimated by multiplying the maximum thrust force of the motor (F) by the total length of the axle. The Factor of Safety (FOS) was set to 2 as defined in the customer requirements. The maximum thrust of a J500 motor is 788 N. Plugging these numbers into equations 1 and 2 yielded preliminary minimum diameter values of 0.358 in and 1.735 in respectively. Thus, the minimum diameter of the center axle was set to 1.75 inches as a starting point of the design.

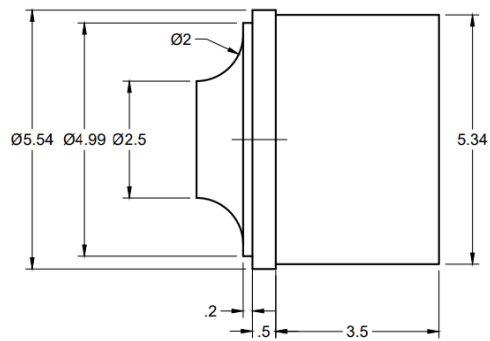
To allow for the system to be assembled, the end caps that hold the center axle into the neighboring body tubes need to be removable so the external orientation system can be assembled. Both ends of the axle are threaded and screw into the end caps to allow for the assembly.



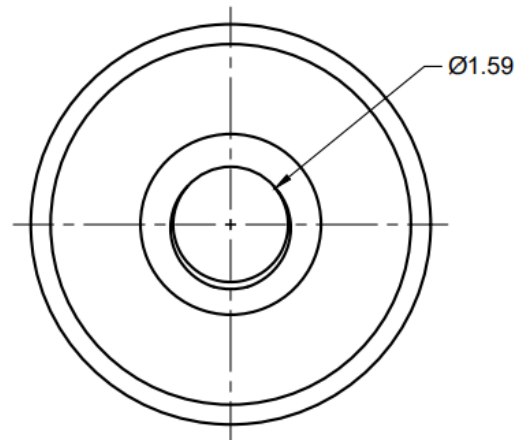
Side View of End Caps



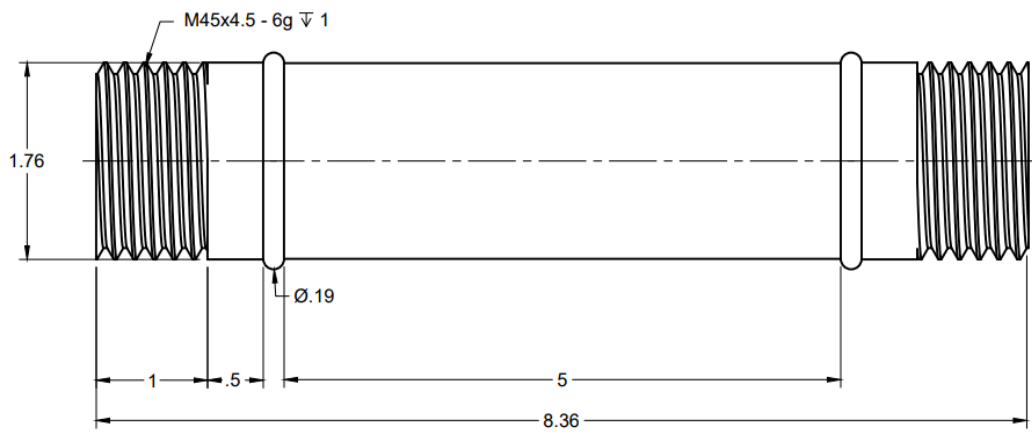
Isometric View of End Caps



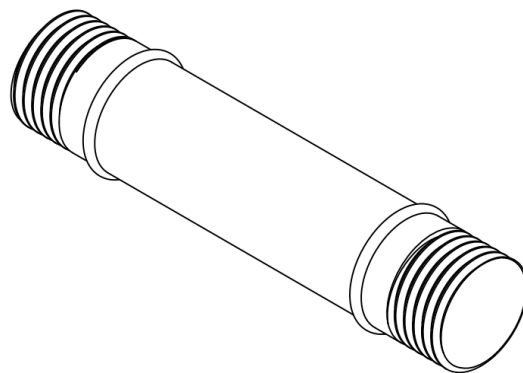
Side View of End Caps



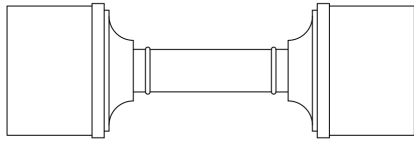
Top View of End Caps



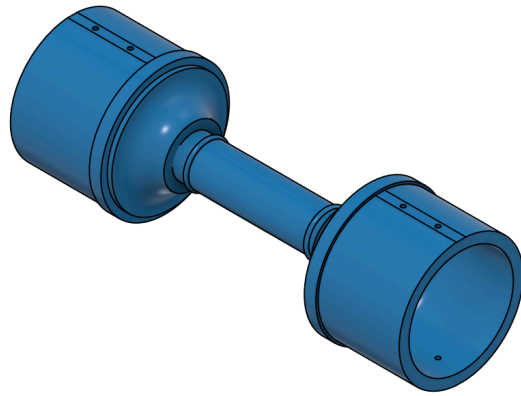
Front View of Support Axle



Isometric View of Support Axle

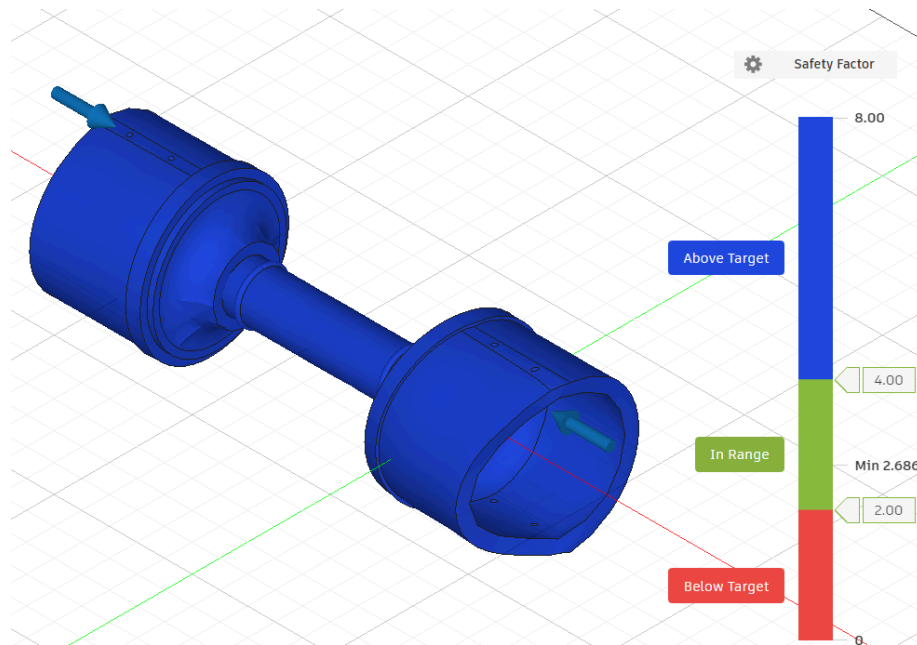


Front View of Assembly



Isometric View of Assembly

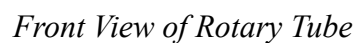
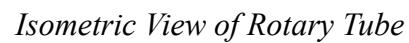
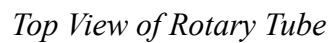
With the support structure modeled in Fusion, finite element analysis (FEA) was used to analyze the structure in order to better refine the preliminary hand calculations and test whether the concept was supported by computer simulations. For the study, compressive loads were applied to the end caps totaling 788 N (the maximum thrust the rocket will experience). The simulated results are shown below.



Compressive Simulation Results

For the compressive test, the FOS of the design falls mostly in the 4-8 range, with a small portion dropping to 2.686, clearly showing the design's capacity to handle the compressive loads

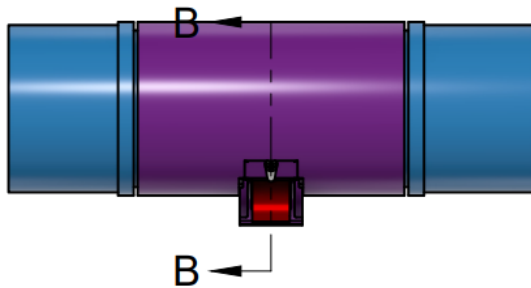
After the verification of the support structure legitimized the design concept, attention could be turned to the external orientation system. Two sets of three bearing wheels will connect to the rotating body tube, holding it firmly in place along the center support axle while allowing smooth and unhindered rotation.



By choosing to close both ends of the rocket to a potential payload ejection, the only remaining option for the team was to create a hatch door in the side of the rotating body tube. The least complex system the team can attack is to make the hatch door open using gravity. Thanks to the orientation system, this is a viable option because the hatch can always be returned to an acceptable angle that allows the weight of the payload to force the door open. A big benefit of this system is its functionality in the limited space between the rotating body tube and internal support axle. More complicated systems simply do not have the space to fit within the assembly.

The mechanism to achieve a gravity fed ejection is simple: attach the payload to the hatch door with a series of walls to keep it contained during flight. This centralizes the payload weight over the hatch door. Hold the hatch door closed with a servo motor to prevent a premature deployment of the rover. When the rocket lands, the servo motor will actuate 90 degrees, releasing the hatch with the rover to a downward angle. Finally, the inside of the hatch must be leveled to create a ramp by which the rover can exit the launch vehicle when the hatch opens.

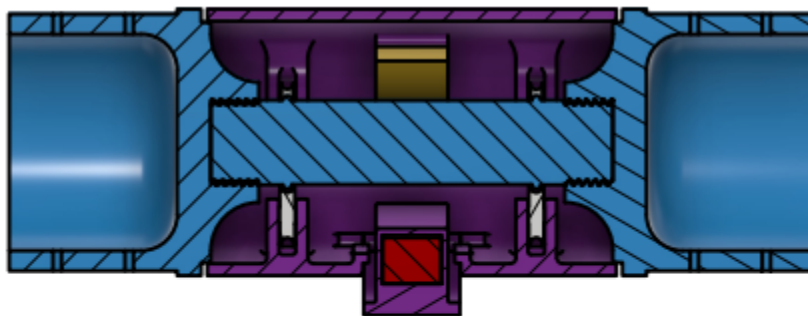
This concept is demonstrated in the assembly figures below. Purple still shows the rotating components and blue still shows fixed, but the red now demonstrates where the payload will sit in the assembly. Inside the tube are a few green and yellow parts to represent the necessary electronics and battery packs used to control the servo motor.



Front View of Rotating Assembly



Section View of Rotating Assembly



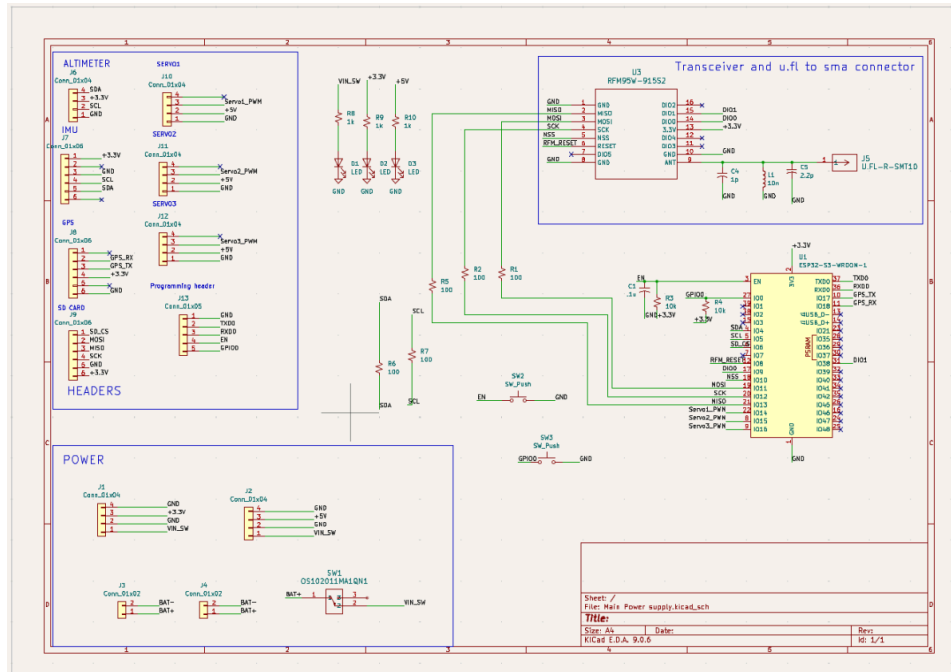
Section View of Rotating Assembly

Electronic Systems

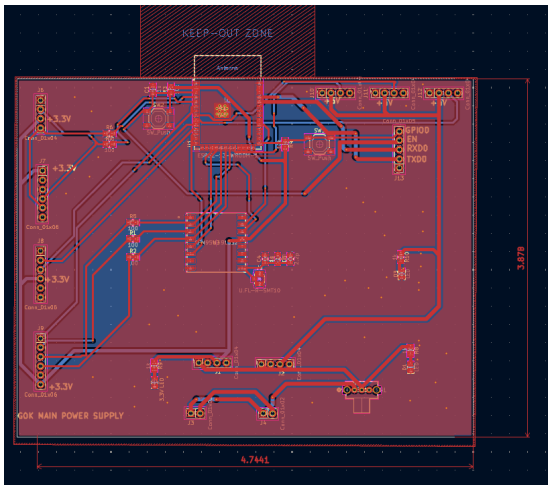
For the following figures, the first is the schematic, the second is the PCB design, and the third is a 3D rendering of the PCB. The dimensions are measured in inches. The size of the actual boards will approximately be equal to the designed PCB. Even if the board were to be redesigned due to errors, the size of the board would not change. Most of the schematics were based on an existing development board, many which have an available trace design that the team referenced. To determine if the measurements for our sensors are practical, we have the development kit values to compare data. Most of the boards have LEDs for voltage indication. Voltage regulation remains within allowable tolerances under worst-case load conditions. The predicted battery runtime exceeds mission duration with appropriate safety margin.

Main Power Supply

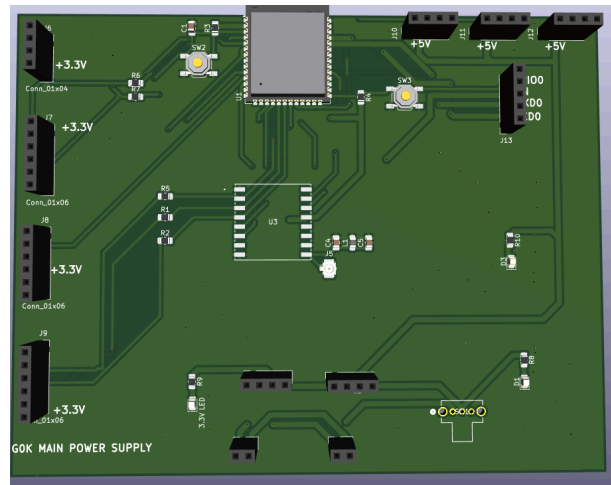
The figures depicted below are the main power supply unit. The board takes in 7.4V from a Li-Ion battery and the regulators adjust the voltage to 3.3V and 5V, voltages that the peripherals need. Most of the other boards will be situated on the headers on the power supply, reducing the total amount of space required to house the electronics. A few LEDs were utilized to determine if the board and the regulators were receiving voltage. This PCB utilizes an ESP32 module to act as the brain of the electronic system. It will take information from the sensors and dictate actions such as moving servos or writing to the microSD card. One of the headers is designated for programming the ESP32, in which we will upload the code.



Schematic of Main Power Supply



PCB Design of Main Power Supply



3D PCB Render of Main Power Supply

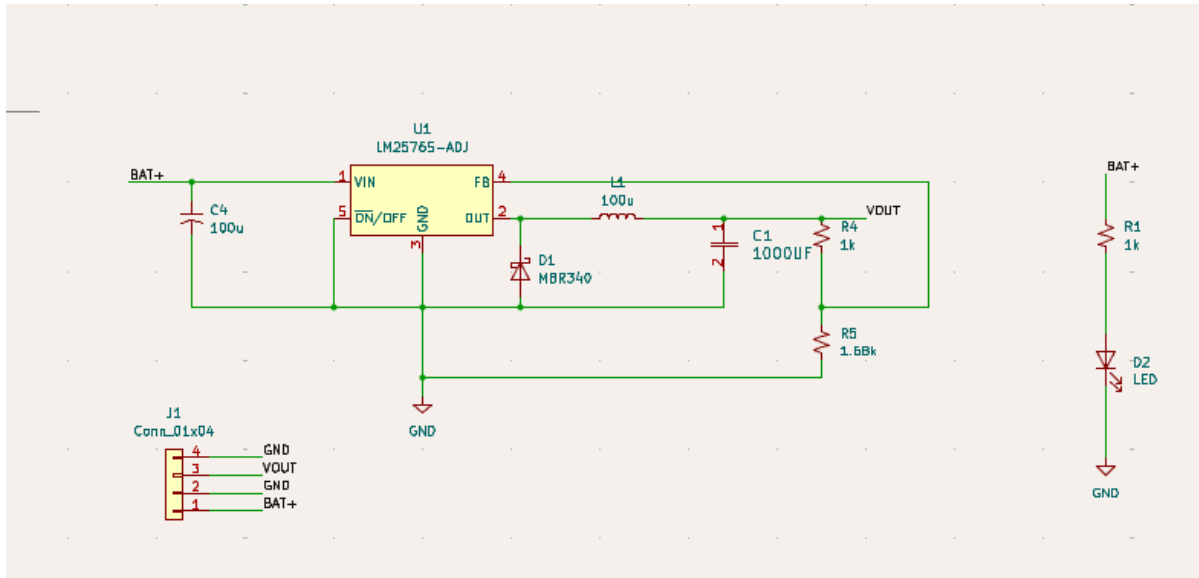
The design is modeled using a power budget and worst-case transient analysis. Average and peak current are estimated for each load (ESP32, LoRa TX bursts, sensors, and especially servo startup/stall spikes). Regulator stability and output ripple are evaluated using datasheet equations for inductor selection, output capacitor ESR limits, and diode ratings. Critical analysis focuses on brownout risk during servo actuation, regulator thermal rise, and voltage droop under

pulse loading; these are validated through bench tests using step-load changes (servo activation) while monitoring 3.3 V and 5 V rails with an oscilloscope.

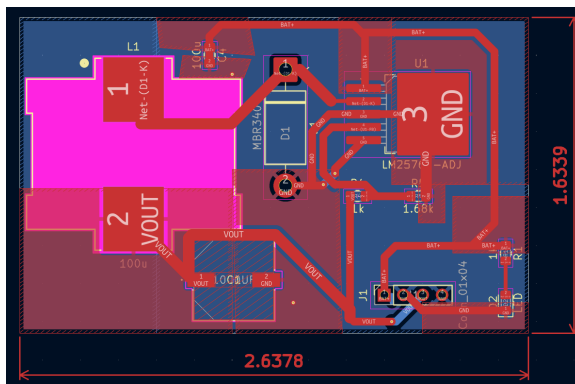
Key parameters include battery voltage (7.4 V for 2S), regulator output voltages (3.3 V logic rail, 5.0 V servo rail), continuous current capability (0.3–1.0 A for logic; 2–3 A peak for servo rail), output ripple, and thermal limits. Missing parameters are usually the true peak current waveform of the SG92R servos under the actual mechanism load and the effective output capacitance under bias; these are determined by measuring inrush and stall current with a current probe and verifying voltage sag on the rail.

Voltage Regulators for Main Power System

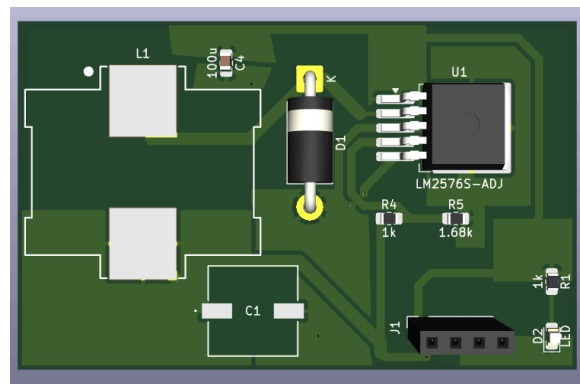
The figures depicted below are the voltage regulator on the main power supply unit. This board utilizes the LM2576S-ADJ regulator to take in the 7.4V from the battery and step down to fixed output voltages of 3.3 V and 5 V. The design of this regulator was based on the schematic in the LM2576 datasheet. The values of the resistors can be changed to control the output voltage value. The voltages can be measured to determine if the system is functioning properly. A LED was placed to determine if the system is getting power. The original design called for a 100 μ H inductor, shown as the biggest component in the figures below, because the team planned to run all the servos through this regulator. During the design, the team decided to separate the servo sections, decreasing the load that the inductor would need to be able to handle. The team decided to still use this inductor because the decision was made when the PCBs and parts were shipping.



Schematic of Voltage Regulator



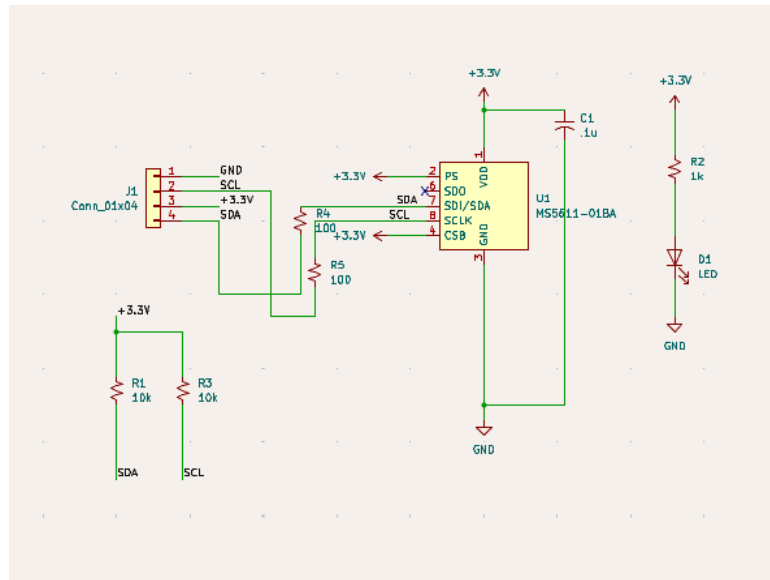
PCB Design of Voltage Regulator



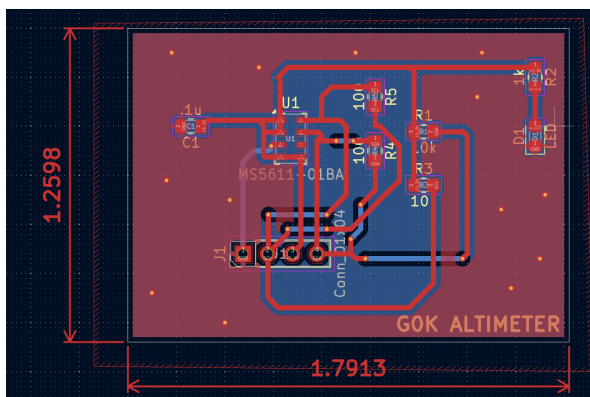
3D PCB Render of Voltage Regulator

Altimeter

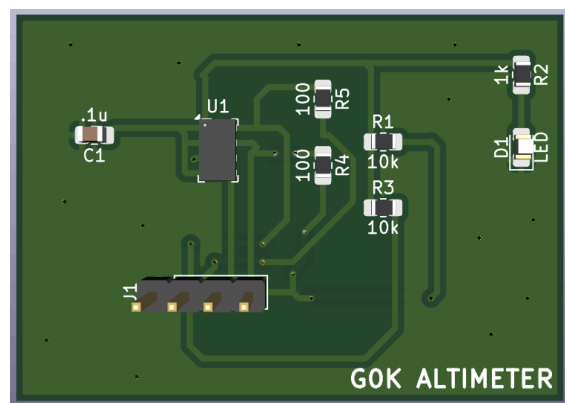
The figures depicted below are the altimeters. This board uses the MS5611-01BA to measure atmospheric pressure to determine the altitude for the payload deployment event. The altimeter's design was referenced from a development board. The altimeter's data can be compared to the development board to determine correctness. The altimeter is one of the more simpler designs, as it only needed the MS5611-01BA to be powered and the information transferred to the main board. The altimeter will be constantly polled for the altitude. A LED was placed to determine if the system is getting power.



Schematic of Altimeter



PCB Design of Altimeter



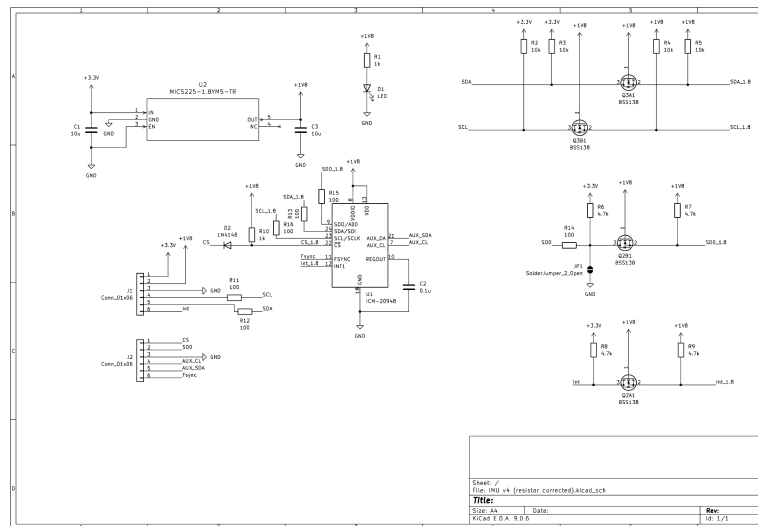
3D PCB Render of Altimeter

Parameters include pressure range, resolution/noise, sampling rate, temperature range, and vent hole geometry (area, placement). A missing parameter that commonly matters is the electronics bay “pressure time constant” driven by vent size and volume; this is determined experimentally by step-changing ambient pressure (or quick vertical movement) and observing sensor lag.

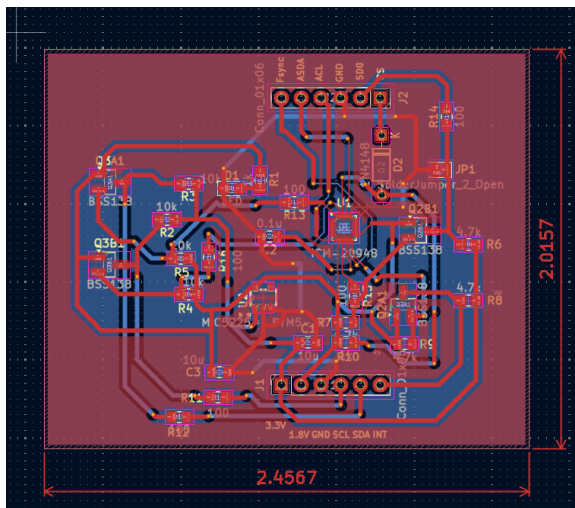
Inertial Measuring Unit (IMU)

The figures depicted below are the inertial measurement unit (IMU). This board uses the ICM-20948, which has a 3-axis gyroscope, 3-axis accelerometer, and 3-axis compass. The IMU’s data will be used in conjunction with the altimeter for the payload deployment event. The

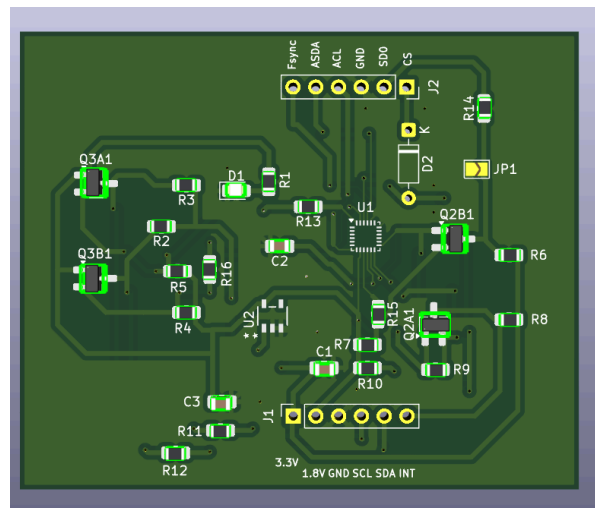
IMU's design was referenced from the SparkFun's and Adafruit's development board. The IMU's data can be compared to the development board for accuracy. A LED was placed to determine if the system is getting power.



Schematic of IMU



PCB Design of IMU



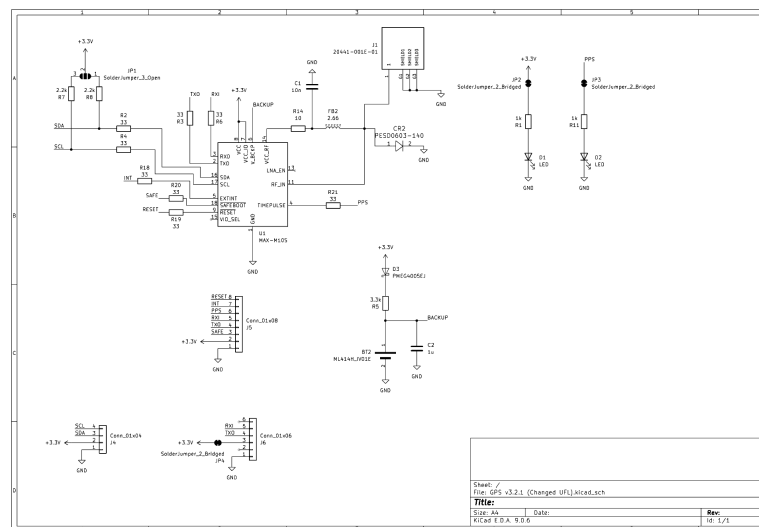
3D PCB Render of IMU

GPS

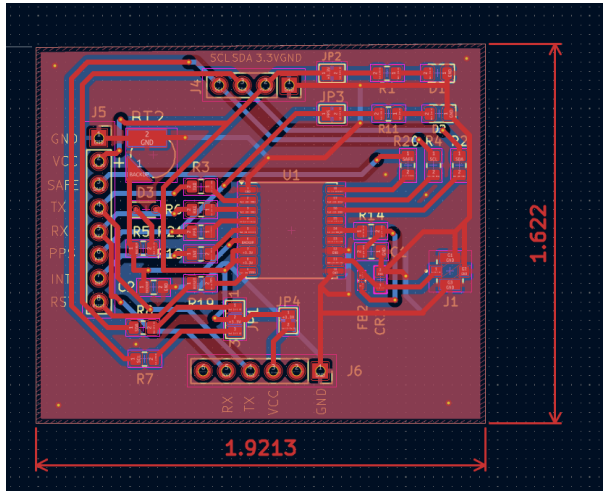
The figures depicted below illustrate the GPS subsystem used in the PLEIADES avionics system. The board is built around the u-blox MAX-M10S module, which utilizes Global Navigation Satellite System (GNSS) signals to determine position, altitude, and velocity. The

design references SparkFun’s development board to ensure proper power filtering, antenna routing, and signal integrity. The custom board can be tested alongside the SparkFun module to compare accuracy, satellite acquisition time, and overall performance. Two onboard LEDs are included—one to indicate power and another to confirm a valid satellite fix—providing quick visual verification of system status during testing and pre-launch preparation.

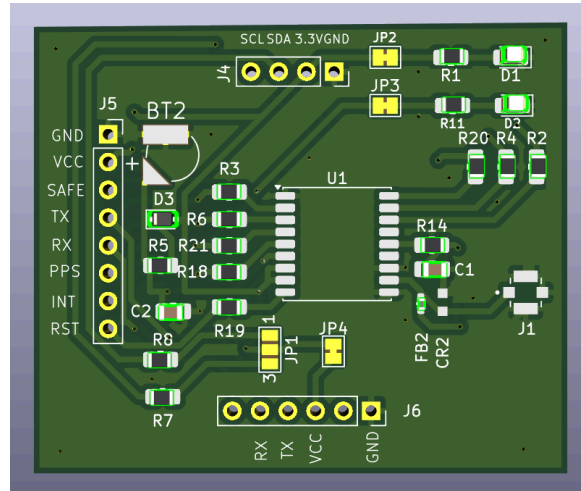
A concern identified by the team is whether the GPS module will maintain reliable satellite lock while housed inside the electronics bay. GNSS signals are relatively weak and can be attenuated by certain structural materials. To address this, the antenna is positioned to face upward with minimal obstruction, and an active patch antenna can be used to improve sensitivity if needed. Ground testing with the electronics fully enclosed will verify signal strength and acquisition reliability prior to flight integration.



Schematic of GPS



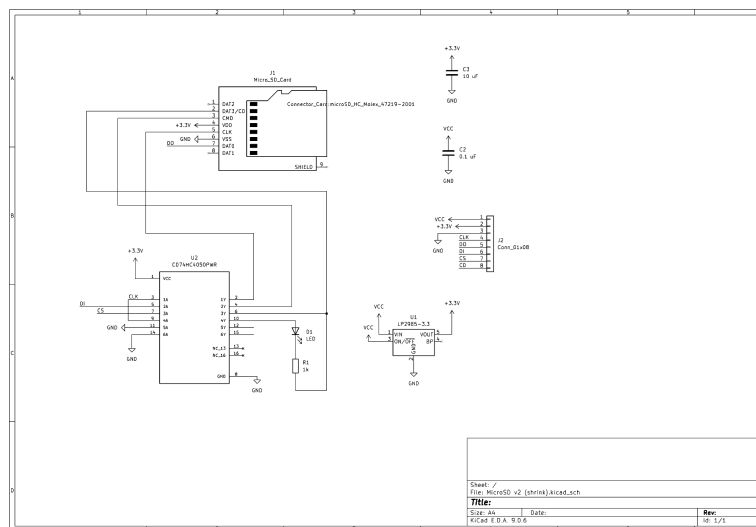
PCB Design of GPS



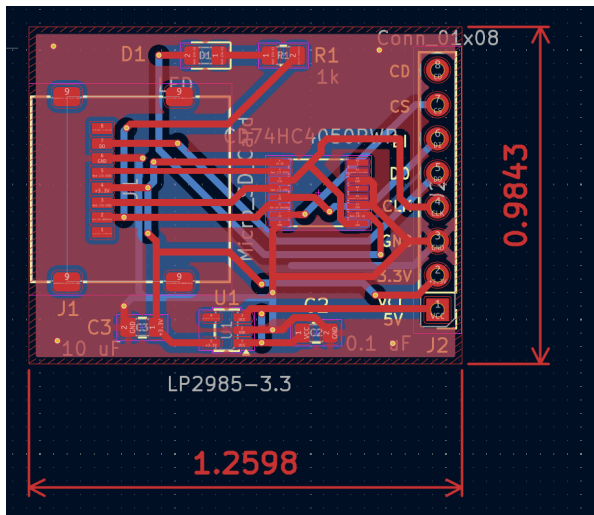
3D PCB Render of GPS

Data Storage System

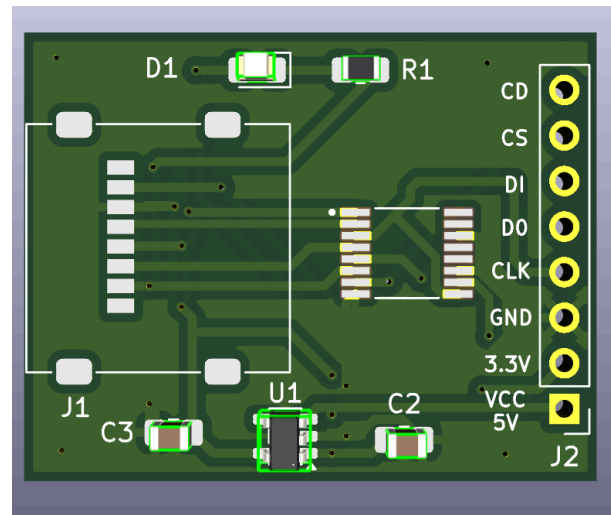
The figures depicted below illustrate the MicroSD card board that will be integrated with the main power supply board to provide onboard data storage for the PLEIADES avionics system. The purpose of this subsystem is to record high-resolution flight data locally during all phases of flight, ensuring that critical information is preserved even if telemetry transmission to the ground station is interrupted.



Schematic of MicroSD Card



PCB Design of MicroSD Card



3D PCB Render of MicroSD Card

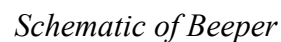
Principles include digital storage reliability, SPI timing, and write-latency management. MicroSD cards have variable write times, so buffering and scheduling matter. Modeling includes estimating data rate (bytes/sec), and maximum write latency to ensure the system doesn't block critical flight tasks. Critical analysis checks that SD writes occur in bursts and that telemetry/deployment tasks remain responsive. Parameters include SPI clock rate, log rate, record size, and card storage. Missing parameters are the worst-case write latency of the specific card; this is found by bench testing with representative data logging rates.

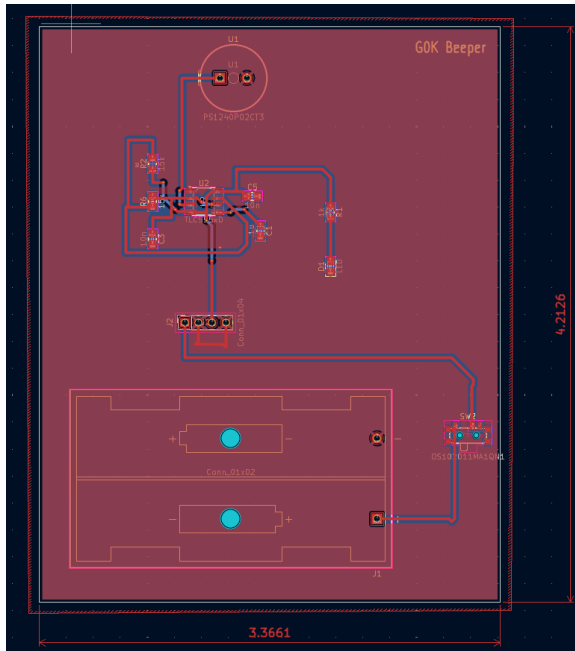
If results align, the SD logger maintains continuous records without corrupting files and without causing missed deployment thresholds. If not, logging must be reduced, buffered more aggressively, or moved to lower priority.

Payload Recovery System

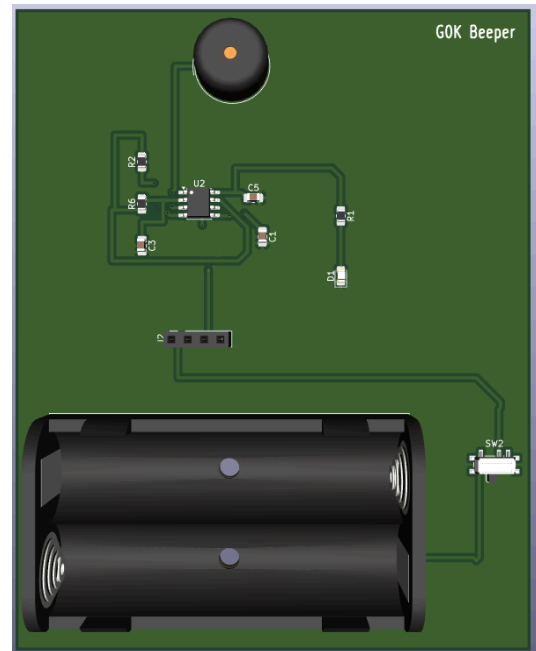
The figures depicted below illustrate the buzzer subsystem that will be installed on each payload to assist with post-flight recovery. Unlike the main avionics system, which is responsible for sensing, control, and telemetry, the buzzer module is designed as a simple, self-contained recovery aid that operates independently once activated.

The buzzer system is implemented using a 555 timer configured in a timing mode that activates the piezoelectric buzzer after a predetermined delay once power is applied. This delay prevents the buzzer from sounding immediately during flight and instead ensures activation after descent and landing. Once triggered, the buzzer emits a continuous audible tone that persists until the battery is depleted or manually switched off. The simplicity of the 555-based circuit reduces failure points and eliminates dependence on sensor inputs, microcontrollers, or communication links.





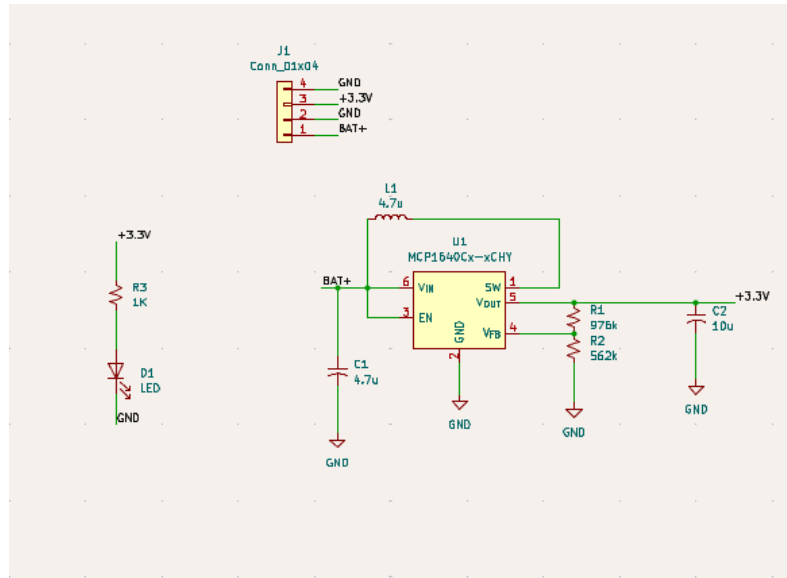
PCB Design of Beeper



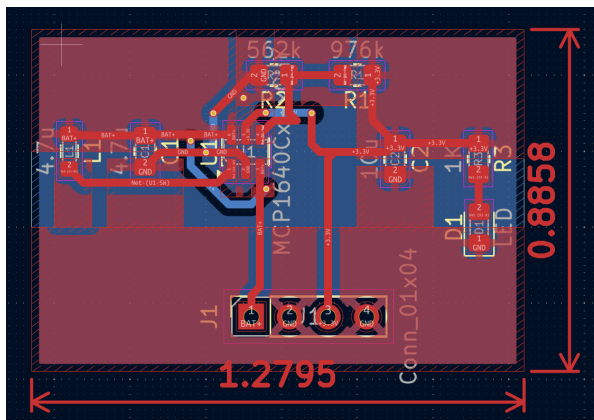
3D PCB Render of Beeper

Low Amperage Voltage Regulators

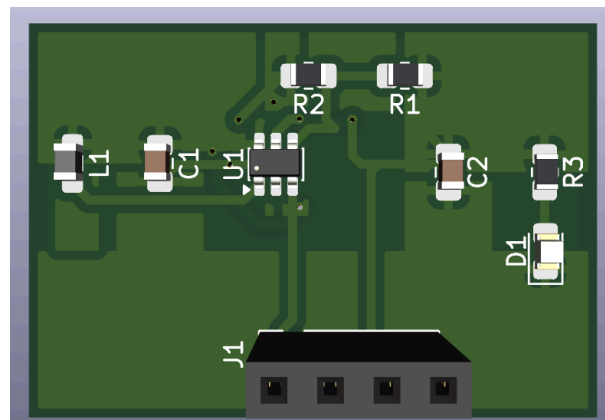
The figures depicted below illustrate the low-amperage voltage regulator circuit used to power the standalone beeper recovery systems installed within each payload. Unlike the primary avionics power system, which must support high-current components such as servos and radio modules, the payload recovery system requires only a stable, low-current supply to operate the 555 timer and piezo buzzer. As a result, a compact linear voltage regulator is sufficient for this application.



Schematic of Voltage Regulator for Beeper



PCB Design for Voltage Regulator for Beeper



3D PCB Render of Voltage Regulator for Beeper

Ground Station

The figures depicted below illustrate the ground station subsystem, which serves as the receiving and visualization interface for the rocket's telemetry system. The ground station is designed to receive wireless flight data transmitted from the onboard LoRa transceiver and forward that information to a laptop through a USB connection. Once connected, the data can be viewed in a serial terminal application such as PuTTY and processed for real-time visualization, including GPS tracking through platforms such as Google Maps.



— 1 —



0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99

output → parsing into terminal/Google Maps. Critical analysis includes ensuring USB serial doesn't drop data and that mapping updates occur smoothly without blocking the receiver loop.

Parameters include USB serial baud rate, packet rate, parsing latency, and antenna configuration at the ground station. Missing parameters are typically environmental RF noise and laptop-side parsing performance; those are determined by field testing. If results align, telemetry appears continuously in PuTTY and the GPS points update on the map with minimal delay. If not, either RF packet loss is occurring or the parsing/USB settings are mismatched to the telemetry rate.

Prototyping and Testing

Following the design and analysis of the individual and composite rocket systems, the team will enter the prototyping phase. Prototyping allows the collection of real data for each subsystem to verify the results of computer analyses. This is an essential part of the engineering process that helps to determine the validity of a design. Computer simulations are a tool for demonstrating the preliminary feasibility of a design, but physical tests are necessary to prove it will function as intended. The team will work in phases when creating prototypes: partial prototyping and full-scale prototyping.

Partial prototyping begins by decomposing the rocket into its individual systems and testing the basic functionality of each. Each of a system's primary functions is tested independently to showcase proof of concept. The acceptable numerical ranges for testing results were specified in the design and analysis stage, and the prototype should be capable of delivering on these ranges. As each system is verified, the team will begin connecting systems via integration testing. Integration testing ensures that systems interact as intended and continue to fulfill their primary functions. This project's main subsystems are the nosecone deployment system, the in-flight payload deployment system, the ground payload deployment system, the airframe, and the electronics system.

Three tests are required for the nosecone payload: that the parachute will deploy when the servo motor is activated, that the parachute is of sufficient size, and that the nosecone can properly detach from the airframe. The servo motor deployment will be tested from various orientations to determine whether an angle exists that inhibits the parachute deployment. The nosecone will also be test-dropped with an open parachute to measure its descent rate. The target landing speed is ft/s to ensure a safe landing. An integration test will then be performed during which the electronics in the nosecone will open the hatch and deploy the parachute during a free-fall from the top of the Libra parking garage. Finally, the nosecone friction fit will be tested under the loading conditions it will experience while in flight, and the tolerances will be adjusted as needed.

For the in-flight payload, the team will test to ensure that the payload is properly released from the body tube by the actuating servo motor. The test will be performed similarly to the nosecone test, checking for any errant orientations. The second test of the in-flight payload is of its parachute. This test will be performed identically to the nosecone payload's chute, and again, the team will ensure that the landing speed is within the range of 15-20 ft/s.

The ground payload deployment subsystems are the orientation and deployment system. For the orientation system, the rotating outer shell will be set to various angles to test the system's ability to return to the stable point. The body tube section with the ground payload will then be placed on the ground, and the electronics will be triggered to test the deployment. This will be the simplest test, as the pass condition is just that the payload successfully exits the rocket. Integration testing can include spinning the rotating outer shell, allowing it to come to rest, and triggering the deployment sequence to ensure the payload is properly oriented upon deployment.

There are multiple functions within the electronics systems, ranging from sensors and data collection to payload deployment and control systems. The team will begin with the power distribution system, validating 3.3V and 5V rails and regulators under load. The integration system between the MCU, sensors, and laptop will be tested to ensure that the software communicates with the sensors and vice versa. The sensors themselves can then be tested by comparing their measurements to those of COTS devices, such as air pressure or acceleration measurements. The team has purchased several "dev kits" that will act as the baseline for said measurements. Last is verifying the servo activation logic and writing software to activate servos for payload deployment, which can be tested through dry runs to ensure functionality.

While partial and analytical prototyping are a necessary step toward crafting a complete design, they do not alone satisfy the requirements of this project. A full-scale prototype is necessary to demonstrate that the design can satisfy the agreed-upon engineering requirements. The full-scale prototype will be launched, and the performance of all systems will be monitored. The test will be deemed a success if the rocket launches to the desired altitude and lands safely, the electronic sensors record values within the set tolerances, and all payloads are deployed properly and at the correct time.

Development Planning

The team is pursuing a fully integrated, high-powered rocket capable of deploying three payloads at distinct heights (at apogee, during descent, and upon landing) while capturing accurate, real-time flight data. The aim is to establish a reusable foundation that future UCF rocket teams can expand upon for advanced research missions involving ascent, descent, and ground-based payload release operations.

Four key objectives define the project's success. First, the payloads must be deployed properly from the rocket at the desired times. The nosecone will be deployed at apogee, a simulated cubesat will be deployed during descent, and a simulated rover will be deployed after landing. Additionally, the safe landing and recovery of the rocket and all its payloads is paramount. If the rocket does not land with a safe descent rate, it cannot be considered a successful launch because it endangers people at the launch site. As the payloads simulate a launch that involves measurement tools to collect data, they must also be recoverable to fulfill design objectives. Finally, the rocket will use custom-designed sensors to report real-time telemetry data to a ground-based tracking station. The data must be collected successfully and accurately within a certain tolerance (when compared to COTS products).

Most of the components of the airframe will be purchased, such as the body tube, couplers, motor casing, and motor. A few structural elements, like the fins, bulkheads, and centering rings, need to be manufactured to help reduce costs. Most of the structural components of the payload deployment mechanisms will be manufactured using 3D printing. Electronically, a combination of COTS components and fabricated sensors will be used for redundancy. Development kits were ordered for the Altimeter, Inertial measuring unit, GPS modules, and a breakout board for the RF module. These parts are selected to reduce development risk and ensure known performance characteristics. Fabricated components primarily include custom PCBs like the main power supply, voltage regulators, and ground station. Sensors will also be fabricated to satisfy ECE requirements. Custom fabrication allows the team to optimize system layout, reduce wiring complexity, and tailor payload deployment mechanisms to mission requirements while leveraging reliable purchased components where possible.

When ordering components, “long lead time” parts need to be identified and planned for to prevent slowdowns in project development. The primary “long lead time” items include custom-fabricated PCBs, specialty sensors, and telemetry modules. PCB fabrication and assembly are critical scheduling factors because design revisions, ordering, and testing cycles can extend several weeks. Additionally, certain avionics components and deployment hardware may experience supply chain variability, requiring early procurement to avoid delays in subsystem integration and testing.

Moving forward, the team will enter the prototyping and testing phase discussed in the section above. System schematics and PCB layouts must be finalized to allow for the fabrication and assembly of avionics boards. Once hardware is received, bench-level validation of power regulation, sensing, and deployment logic must be completed before subsystem integration. Integration testing then leads into full-system ground testing and pre-flight validation. Any delays in PCB completion, deployment testing, or avionics integration directly affect the launch-readiness timeline, making these tasks central to the development schedule. Payload deployment systems will be tested simultaneously, but because these systems rely on 3D printing for manufacturing, they can be more easily modified and adjusted than the electronics systems. With partial prototyping and testing complete, the team will move toward a full-scale prototype flight test on March 14th (weather may delay this to April).

On a broad scale, the team is split into two groups: Electrical/Computer (ECE) and Mechanical/Aerospace (MAE) Engineers. Responsibilities are assigned according to each discipline’s technical expertise to ensure clear ownership and efficient subsystem development. The Electrical Engineering team members are responsible for the electrical hardware implementation, power distribution, voltage regulation, electrical integration, and hardware-level validation of avionics components. The Computer Engineering lead is responsible for embedded software and firmware development for the avionics board, including sensor interfacing, deployment logic implementation, communication protocols, and overall control system programming. The Mechanical and Aerospace Engineering team members manage structural integration, payload design, deployment mechanisms, recovery systems, propulsion, and overall flight operations planning. Zooming in, there are engineers assigned to each subsystem (nosecone deployment, in-flight deployment, ground deployment, and flight). The engineers

responsible for their respective subsystems will lead component and subsystem testing. Collaboration across all disciplines occurs during integration, testing, and pre-flight validation to ensure the full rocket system operates reliably as a unified platform.